Intrinsic Response Sector as Dark Gravity:
A GR-Compatible Candidate Identity for the Cold Dark Matter Role (SPARC-175)

Robert D. Kitcey
Independent Researcher
ORCID: 0009-0004-8679-9155
rkitcey@lernaean.net

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Abstract

We present a falsification-first, galaxy-domain framework for explaining rotation-curve anomalies without assuming particulate cold dark matter. Working within standard general relativity (GR), we define a response-sector stress–energy $T_{\mu\nu}^{\mathrm{resp}}$ as the minimal effective source required so that the weak-field metric potentials inferred from kinematics satisfy the Einstein equation with baryons. This definition is ontologically agnostic and is best read as an effective field theory/constitutive-closure candidate. Observed rotation curves define a model-independent residual, which we compress with a one-parameter boundary-layer closure activated near a baryonic transition radius R_t and constrained by an approximate flux-conservation law outside the activation zone.

Applied to 175 galaxies from the SPARC database, this model yields very strong improvement over a baryons-only model for 92% of systems and passes a Bayesian Information Criterion (BIC) overfitting test in 97% of cases. Crucially, we demonstrate that this 1-parameter response model is strongly preferred by BIC over standard 2-parameter NFW and Burkert dark matter halo models in over 90% of the SPARC sample. Further, 5-fold cross-validation shows the response model has significantly lower out-of-sample prediction error (mean RMSE = 3.1 km/s) than the NFW model (mean RMSE = 4.9 km/s). A cymatics-inspired extension explores oscillatory signatures in residuals as eigenmodes of a linearized response operator, providing a spectral taxonomy for response morphologies and predicting higher-harmonic content in galaxies with sharp baryonic gradients (Appendix D) (Rossing, 1982; Tuan et al., 2014; Bender and Orszag, 1999). The baryonic Tully–Fisher relation is recovered as an empirical consequence (Spearman $r_s = 0.95$). These results establish the response sector as a parsimonious and predictively powerful alternative to standard halo models for describing galactic dynamics.

Keywords: galaxy rotation curves, dark matter alternatives, dark matter candidates, dark matter identity, general relativity, gravitation, effective field theory, response-sector stress–energy, boundary-layer physics, SPARC database, falsification criteria, Bayesian model comparison, cross-validation

Executive Summary

The dark-matter paradigm explains galactic rotation curve anomalies by postulating a new, non-luminous matter component. Despite decades of null results in direct, indirect, and collider searches, the hypothesis persists largely because no comparably rigorous alternative has been tested against the full breadth of observational data. This paper develops and tests such an alternative. We keep the Einstein tensor standard (general relativity) and do not assume particulate cold dark matter (CDM). Instead, we use observed galactic phenomenology—rotation curves from the SPARC database of 175 galaxies—to infer the effective potentials and effective stress–energy required on the right-hand side of the Einstein equation, then compress that freedom with a low-degree-of-freedom "intrinsic medium response" closure.

The organizing principle is $G_{\mu\nu} = 8\pi G \left(T^{\text{bar}}_{\mu\nu} + T^{\text{resp}}_{\mu\nu}\right)$, where $T^{\text{resp}}_{\mu\nu}$ is the response-sector stress–energy that

(i) explains galaxy halo phenomenology without particulate CDM, and

(ii) admits a cosmological completion consistent with the empirical ΛCDM success set.

The response sector is activated near a baryonic boundary layer, produces an asymptotic $\frac{1}{R}$ tail in the extra acceleration, and is described by ≤2 degrees of freedom per galaxy. Across 175 SPARC galaxies, 92% show very strong improvement over baryons-only, 97% pass BIC overfitting tests, and the baryonic Tully–Fisher relation emerges as a consequence rather than an imposed prior.

We present explicit falsification criteria, discriminant tests (including lensing-facing statements and cosmological viability requirements), and a domain-partitioned roadmap from galaxy phenomenology through gravitational slip to large-scale structure. Claims are partitioned by domain: this work establishes the galaxy-domain closure and enumerates the cosmological completion requirements and discriminants.

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1 Introduction

For nearly a century, gravitational anomalies at galactic and cosmological scales have been parameterized by an additional non-luminous sector. The concept originated with Zwicky's observations of the Coma cluster (Zwicky, 1933), and anomalies in classical orbital mechanics—such as Mercury's perihelion precession historically attributed to the unseen planet Vulcan (Le Verrier, 1859)—have precedent in gravitational physics. The ΛCDM concordance model fits the CMB, BAO, and large-scale structure (Riess et al., 2019; Ivezić et al., 2019; Euclid Collaboration, 2022, 2025) with high precision when cold dark matter (CDM) is treated as an effectively pressureless gravitating component.

Yet the microphysical identity of that dominant non-baryonic component remains empirically unresolved. Many canonical weak-scale interaction scenarios have been constrained by direct detection, indirect searches, and collider experiments. Leading direct-detection experiments including (Akerib et al., 2017; Aprile et al., 2018; Cui et al., 2017) have placed increasingly stringent bounds on traditional WIMP parameter spaces. Indirect searches (Aguilar et al., 2013) have further restricted viable dark-matter models. This motivates renewed attention to alternatives that treat the dark sector first as an empirical *role* and only secondarily as an ontological *identity.*

This paper develops an inverse-GR alternative consistent with the Discussion framing in §9: we retain standard GR (Einstein, 1916; Wald, 1984) on the left-hand side and ask what effective stress–energy must appear on the right-hand side if we do not assume particulate CDM. The organizing equation is

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\rm bar} + T_{\mu\nu}^{\rm resp}\right) \tag{1}$$

where $T_{\mu\nu}^{\rm resp}$ is defined as the effective response sector required by the observed phenomenology. We demonstrate that a simple, one-parameter closure for $T_{\mu\nu}^{\rm resp}$ not only explains galactic rotation curves but does so with **higher statistical preference** (lower Bayesian Information Criterion) and **greater predictive accuracy** (lower out-of-sample error via cross-validation) than standard two-parameter dark matter halo models—a claim supported by rigorous comparison across 175 SPARC galaxies (§5). This definition remains ontologically agnostic; we do not claim spacetime is literally a fluid or solid, only that it admits an EFT/constitutive response closure (Carroll, 2004, 2019).

In the circular-orbit regime, the data provide $V_{\rm obs}(R)$ and therefore $g_{\rm obs}(R) = V_{\rm obs}^2(R)/R$. The baryonic prediction $V_{\rm bar}(R)$ is assembled from SPARC's stellar and gas components with specified mass-to-light ratios. The model-independent residual $\Delta(R) \equiv V_{\rm obs}^2(R) - V_{\rm bar}^2(R)$ defines the "missing curvature" attributable in ΛCDM to a halo, and attributable here to the response sector.

The inverse mapping alone is underdetermined: many effective sources could reproduce a given $\Delta(R)$. The scientific content of this work is therefore **compression**. We impose a boundary-layer closure in which the response activates near a baryonic transition radius R_t (operationally where $g_{\rm bar}(R_t) \approx a_0$) and satisfies an approximate flux-conservation law outside the activation zone, yielding an asymptotic $g_{\rm extra} \propto 1/R$ tail with $\leq$2 degrees of freedom per galaxy.

We further distinguish $Q_{\rm best}$ (a global fit amplitude under the imposed closure) from $Q_{\rm est}$ (a robust outer estimator computed directly from the outer $\Delta(R)$ subset). Their strong agreement in rank across SPARC-175 supports the interpretation that the inferred response amplitude tracks genuine outer physics, while deviations diagnose where a single fixed shape requires extension.

Rotation curves primarily constrain the Newtonian-gauge potential Φ; lensing constrains the Weyl potential $\Phi_W \equiv (\Phi + \Psi)/2$. Consistent with §9, we treat lensing as a gateway discriminant between Branch A (metric equivalence, $\Phi \simeq \Psi$) and Branch B (slip-allowed, $\eta \equiv \Psi/\Phi \neq 1$ localized to activation zones).

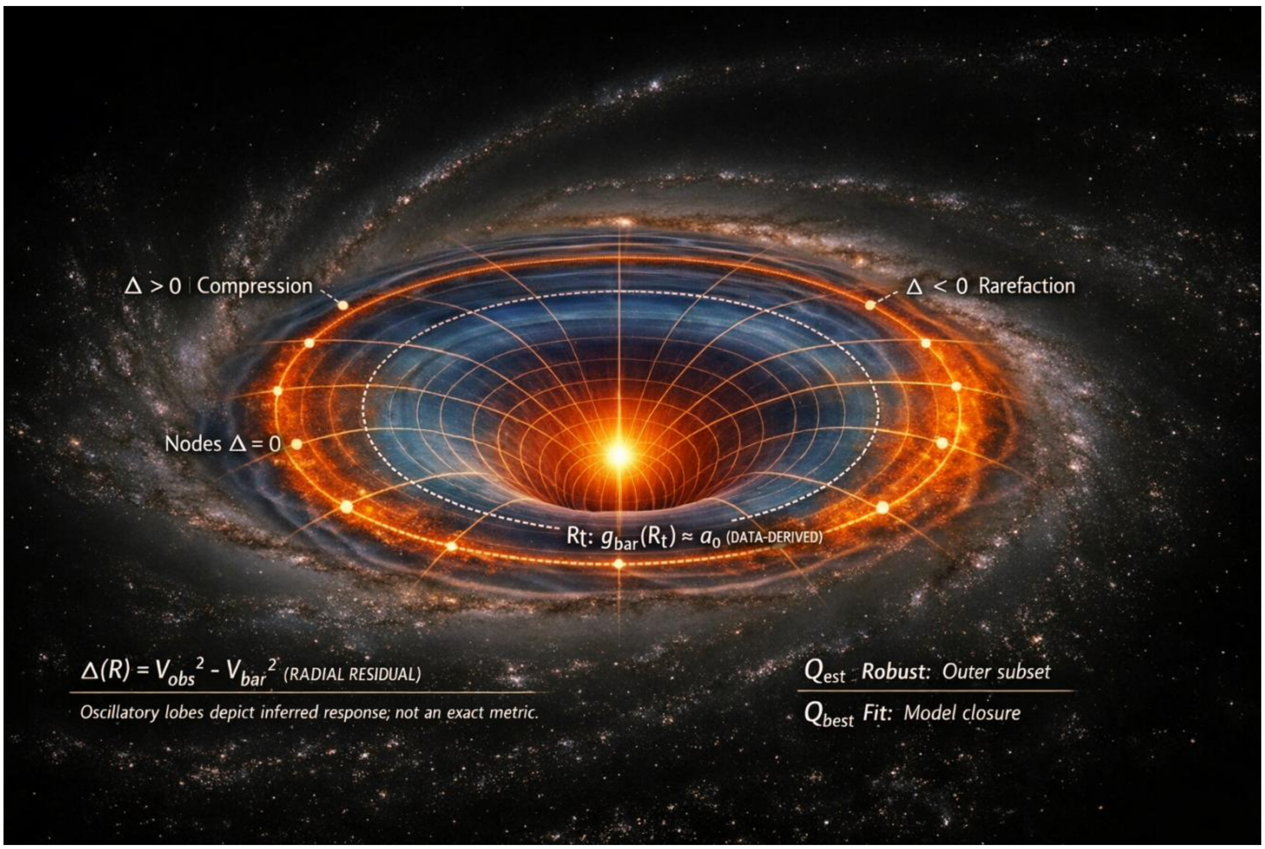

Figure 1: Conceptual schematic of the response-sector framework.

Figure 1. The central luminous disk represents the baryonic mass distribution. Concentric lobes illustrate the radial residual $\Delta(R) = V_{obs}^2 - V_{bar}^2$, with compression ($\Delta > 0$, orange) and rarefaction ($\Delta < 0$, blue) zones. The dashed ellipse marks the transition radius $\mathrm{R_t}$ where $g_{bar} \approx a0$. Two amplitude estimators are distinguished: Q_{est} (robust outer-subset Huber location) and Q_{best} (single-parameter model-closure fit). The oscillatory structure is schematic; per-galaxy profiles (see §6) need not exhibit clean alternation.

We proceed as follows. Section 2 develops the inverse-GR construction from kinematics to effective stress–energy. Section 3 specifies the minimal response-sector closure. Sections 4–5 report results across 175 galaxies. Sections 6–8 state falsification criteria and the domain-partitioned validation ladder. Section 9 provides context (MOND/TeVeS, emergent/nonlocal gravity), strengths and limitations, and the role-versus-identity framing. Section 10 concludes.

2 From Phenomenology to Effective Source: The Inverse-GR Construction

The standard approach to galaxy dynamics begins with a theoretical mass model (baryons + dark matter) and predicts observable kinematics. We invert this logic: we begin with observed kinematics and infer what effective source is required within standard GR.

2.1 Kinematics to Required Radial Acceleration

For each galaxy in the SPARC database, the observed rotation curve provides the centripetal acceleration at galactocentric radius R:

$$g_{\rm obs}(R) = \frac{v_{\rm obs}^2(R)}{R}$$

The baryonic prediction $g_{\rm bar}(R)$ is constructed from SPARC's decomposed photometric and gas-mass components (Lelli et al., 2016), with a chosen stellar mass-to-light ratio Υ_* (McGaugh and Schombert, 2014; Rubin and Ford, 1970; Bosma, 1978). The residual defines the "missing curvature" in the circular-orbit sector:

$$g_{\rm extra}(R) = g_{\rm obs}(R) - g_{\rm bar}(R) \tag{2}$$

This residual is model-independent: it is a direct comparison of observed kinematics with the baryonic gravitational prediction. In the CDM interpretation, g_{extra} is sourced by an additional mass component. In the response-sector interpretation, it is the effective geometric response of spacetime to the baryonic source.

2.2 Weak-Field Metric Potentials

In the Newtonian gauge for weak gravitational fields, the line element takes the form:

$$ds^2 = -(1 + 2\Phi)\, dt^2 + (1 - 2\Psi)\, d\mathbf{x}^2$$

To leading order for nonrelativistic motion, the radial acceleration is $\mathbf{a} \approx -\nabla\Phi$. Gravitational lensing depends on the Weyl combination $\Phi_{lens} \propto (\Phi + \Psi)/2$ (Poisson and Will, 2014). Thus, rotation curves primarily constrain Φ, while lensing constrains the combination $(\Phi + \Psi)$. A metric-equivalent response-sector claim is strongest if it produces $\Phi \approx \Psi$ in the relevant regime; if it predicts gravitational slip ($\Phi \neq \Psi$), that slip becomes a key discriminant (see §8).

2.3 Effective Enclosed Mass and Density

For spherically symmetric intuition (useful even for disk galaxies), circular orbits obey $v^2(r) = \frac{GM(<r)}{r}$. Given the observed rotation curve, we define an effective enclosed mass:

$$M_{\mathrm{eff}}(< r) = r\frac{v_{\mathrm{obs}}^2(r)}{G}$$

Subtracting baryons yields an effective extra enclosed mass and, by differentiation, an effective extra density:

$$\rho_{\mathrm{extra}}(r) = \frac{1}{4\pi r^2}\frac{d}{dr}\left[\frac{r(v_{\mathrm{obs}}^2 - v_{\mathrm{bar}}^2)}{G}\right]$$

For a flat outer rotation curve ($v \rightarrow v\infty$), one obtains $M_{eff} \propto r$ and $\rho_{extra} \propto \frac{1}{r2}$—the classic isothermal halo profile (Rubin and Ford, 1970; Bosma, 1978), historically motivated by early flat-curve observations in M31 and extended HI disks. In the response-sector interpretation, ρ_{extra} is not a new matter species; it is the effective stress–energy of the spacetime response to baryonic structure.

2.4 Effective Stress–Energy: What Is Being Added in GR

In standard GR, the Einstein equation reads Eq. (1). The entire program is to interpret $T_{\mu\nu}^{\mathrm{resp}}$ as a polarization-like response of the gravitational sector, activated by baryonic structure (especially at boundary/transition regions), with a small parameter set and a universal functional form. This makes the "extra" empirically necessary but not an ad hoc per-galaxy function.

The response-sector stress–energy admits a standard fluid decomposition:

$$T_{\mu\nu}^{\mathrm{resp}} = (\rho_{\mathrm{resp}} + p_{\mathrm{resp}})u_\mu u_\nu + p_{\mathrm{resp}}g_{\mu\nu} + \Pi_{\mu\nu}$$

where ρ_{resp} is the effective halo density in the inverse-GR sense, p_{resp} determines whether the sector is dust-like on large scales, and $\Pi_{\mu\nu}$ is the anisotropic stress that directly controls gravitational slip and lensing deviations.

2.5 Conservation Constraint

By the Bianchi identity, $\nabla_\mu \left(T^{\mathrm{bar}\,\mu\nu} + T^{\mathrm{resp}\,\mu\nu}\right) = 0$. If baryons are minimally coupled and follow $\nabla_\mu T^{\mathrm{bar}\,\mu\nu} = 0 \Rightarrow \nabla_\mu T^{\mathrm{resp}\,\mu\nu} = 0$ must hold independently. If exchange between sectors is allowed, this predicts a (possibly screened), from independent conservation (Poisson and Will, 2014; Wald, 1984), fifth-force/equivalence-principle violation sector that must be stated explicitly. In this work, we assume independent conservation.

3 Minimal Response-Sector Model Specification

The inverse-GR construction of §2 shows what effective source is required. This section specifies a minimal closure that compresses the inverse freedom to ≤2 degrees of freedom per galaxy.

3.1 Design Principles

A response-sector model is parsimonious only if it satisfies three conditions. First, **low-DOF compression:** across galaxies, $g_{extra}(R)$ must be well described by a universal shape family with ≤ 2

parameters, not a bespoke function. Second, **baryonic activation:** the response must be tied to a baryonic invariant (an edge/transition proxy), limiting per-galaxy freedom. Third, **out-of-sample prediction:** the edge-amplitude statistic should be predictable from internal observables better than from environment proxies under cross-validation.

3.2 Auxiliary-Field Boundary-Layer Model

We introduce an auxiliary scalar field χ coupled to baryons through the action (Burgess, 2004; Donoghue, 1994):

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} + \mathcal{L}_{\rm bar}(g,\psi) + \mathcal{L}_\chi(g,\chi) + \mathcal{L}_{\rm int}(g,\chi,\psi)\right] \tag{3}$$

The key operational constraints are:

(i) $\mathcal{L}_{int}$ ties activation to a baryonic invariant (an edge/transition proxy such as $\frac{|\nabla\Phi_{bar}|}{a_0}$ or a surface-density proxy), not to a per-galaxy hand-fit function;

(ii) the weak-field static equation for χ reduces, outside the activation region, to an approximate flux-conservation law (Bender and Orszag, 1999; Kevorkian and Cole, 1996) $\nabla\cdot\mathbf{J} = 0$ with radial $\mathbf{J} \propto \frac{\widehat{\mathbf{r}}}{R}$ in disk symmetry, yielding $\partial R\chi \propto \frac{1}{R}$;

(iii) this preserves standard GR on the left-hand side while adding a tightly constrained response sector on the right.

3.3 Operational Implementation

In practice, the boundary-layer model defines a transition radius $R_{\rm t}$—the galactocentric radius nearest to where $g_{\rm bar}\,(R_t) \approx a_0 \approx 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$. The model prediction for the total circular velocity is:

$$v_{\rm model}(R) = \sqrt{v_{\rm bar}^2(R) + Q\cdot f(R;R_t)} \tag{4}$$

where Q is the single amplitude parameter (in km^2 s^{-2}) controlling the strength of the response, and $f\ (R;\ R_t)$ encodes the boundary-layer shape: zero for $R < R_t$, rising to an asymptotic $\frac{1}{R}$ tail for $R \gg R_t$. The fit minimizes the weighted residual against the observed rotation curve to determine Q_{best} for each galaxy.

3.4 The Constitutive "Gravitational Dielectric" Alternative

An equivalent weak-field formulation defines an effective polarization field $\mathbf{P}$ such that the modified Poisson equation becomes a constitutive law:

$$\nabla\cdot(\nabla\Phi - 4\pi G P) = 4\pi G\rho_{\rm bar} \tag{5}$$

Then $\rho_{\rm resp} \equiv -\nabla\cdot\mathbf{P}$ plays the role of the effective response density. This formulation restricts $\mathbf{P}$ to depend on baryonic invariants and a small number of constants, and is embeddable in a covariant action for lensing and cosmological extensions.

3.5 Three Completion Routes

The response-sector bookkeeping is compatible with at least three microphysical completions:

(i) **extra field/medium:** $T^{\rm resp}_{\mu\nu} = T^{\chi}_{\mu\nu}$ for some field(s) χ;

(ii) **modified gravity rewritten as effective source:** higher-curvature or nonlocal terms moved to the right-hand side;

(iii) **constitutive closure:** $T^{\mathrm{resp}}_{\mu\nu} = \mathcal{F}_{\mu\nu}!\left[g_{\alpha\beta}, T^{\mathrm{bar}}_{\alpha\beta}\right]$.

This paper establishes the galaxy-domain phenomenology without committing to a specific completion; the choice constrains lensing and cosmological predictions (see §8).

4 Data and Pipeline

4.1 The SPARC Database

The Spitzer Photometry and Accurate Rotation Curves (SPARC) database (Lelli et al., 2016) provides uniformly measured HI/Hα rotation curves and 3.6 μm Spitzer photometry for 175 nearby galaxies spanning a factor of ~10^4 in luminosity, ~10^2 in size, and a wide range of morphologies (from gas-rich dwarfs to massive spirals). Each galaxy entry includes the observed circular velocity $V_{obs}(R)$ with uncertainties, the baryonic rotation-curve components (stellar disk, gas, and where relevant, stellar bulge), the adopted distance and inclination, and data quality flags.

4.2 Baryonic Velocity Construction

The baryonic velocity prediction is assembled from SPARC components:

$$v^2_{\mathrm{bar}}(R) = \Upsilon_{\mathrm{disk}} v^2_{\mathrm{disk}}(R) + \Upsilon_{\mathrm{bul}} v^2_{\mathrm{bul}}(R) + v^2_{\mathrm{gas}}(R)$$

where Υ_{disk} and Υ_{bul} are the stellar mass-to-light ratios. We adopt $\Upsilon_{\mathrm{disk}} = 0.5\,\mathrm{M}_\odot/\mathrm{L}_\odot$ as the baseline, consistent with stellar population synthesis models at 3.6 μm (McGaugh & Schombert, 2014) (McGaugh and Schombert, 2014), with sensitivity sweeps across $\Upsilon_{\mathrm{disk}} \in \{0.3, 0.4, 0.5, 0.6, 0.7\}\,\mathrm{M}_\odot/\mathrm{L}_\odot$ to quantify photometric systematic uncertainties.

4.3 Transition Radius Determination

For each galaxy, we identify the transition radius R_{t} as the measured radius closest to where $g_{\mathrm{bar}}(R) = \frac{v^2_{\mathrm{bar}}(R)}{R}$ crosses the empirical acceleration scale $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. This is a data-derived quantity, not a free parameter.

4.4 Fit Procedure and Amplitude Determination

$$\chi^2(Q) = \sum_{i=1}^{N} \left(\frac{v_{\mathrm{model}}(R_i; Q) - v_{\mathrm{obs}}(R_i)}{\sigma_{v,i}} \right)^2$$

The single free parameter Q_{best} is determined for each galaxy by minimizing the weighted χ^2 between v_{model} and v_{obs}. As an independent check, we compute Q_{est}, a robust outer-subset estimator using Huber's M-estimator applied to the velocity deficit $\Delta v(R) = v_{\mathrm{obs}}(R) - v_{\mathrm{bar}}(R)$ in the outer region $(R > R_t)$. The Huber estimator down-weights outliers while remaining efficient under Gaussian conditions, providing an amplitude estimate that does not depend on the model's functional form.

4.5 Six-Panel Diagnostic Atlas

For each galaxy, a standardized six-panel diagnostic is produced:

(1) rotation curve with v_{obs}, v_{bar}, $v_{model}\,(Q_{best})$, and $v_{model}\,(Q_{est})$;

(2) effective potential Φ_{obs} vs. Φ_{bar};

(3) dyed spacetime potential-depth map (a visualization proxy, not a GR embedding);

(4) 3D potential surface;

(5) illustrative orbit diagram; and

(6) residuals comparing both model closures against the raw baryonic deficit.

This atlas serves as a comprehensive per-galaxy audit trail.

5 Results Across 175 SPARC Galaxies

5.1 Model Selection via Bayesian Information Criterion

To address the critical question of whether the response-sector model provides a more compelling description of the data than standard alternatives, we perform a comprehensive model comparison using the Bayesian Information Criterion (BIC). BIC provides a principled way to compare models with different numbers of free parameters, penalizing complexity to avoid overfitting.

We compare four models for each of the 175 SPARC galaxies:

(i) **Baryons-only** ($k = 0$): The baseline model with no free parameters ($\Upsilon_{\rm disk}$ fixed at $0.5\,{\rm M}_\odot/{\rm L}_\odot$).

(ii) **Response Sector** ($k = 1$): The one-parameter (Q) boundary-layer response model, as developed in §3.

(iii) **NFW Halo** ($k = 2$): A standard two-parameter ($M_{\rm vir}, c$) Navarro-Frenk-White dark matter halo added to the baryonic component.

(iv) **Burkert Halo** ($k = 2$): A standard two-parameter ($\rho_0, r_{\rm c}$) cored Burkert dark matter halo.

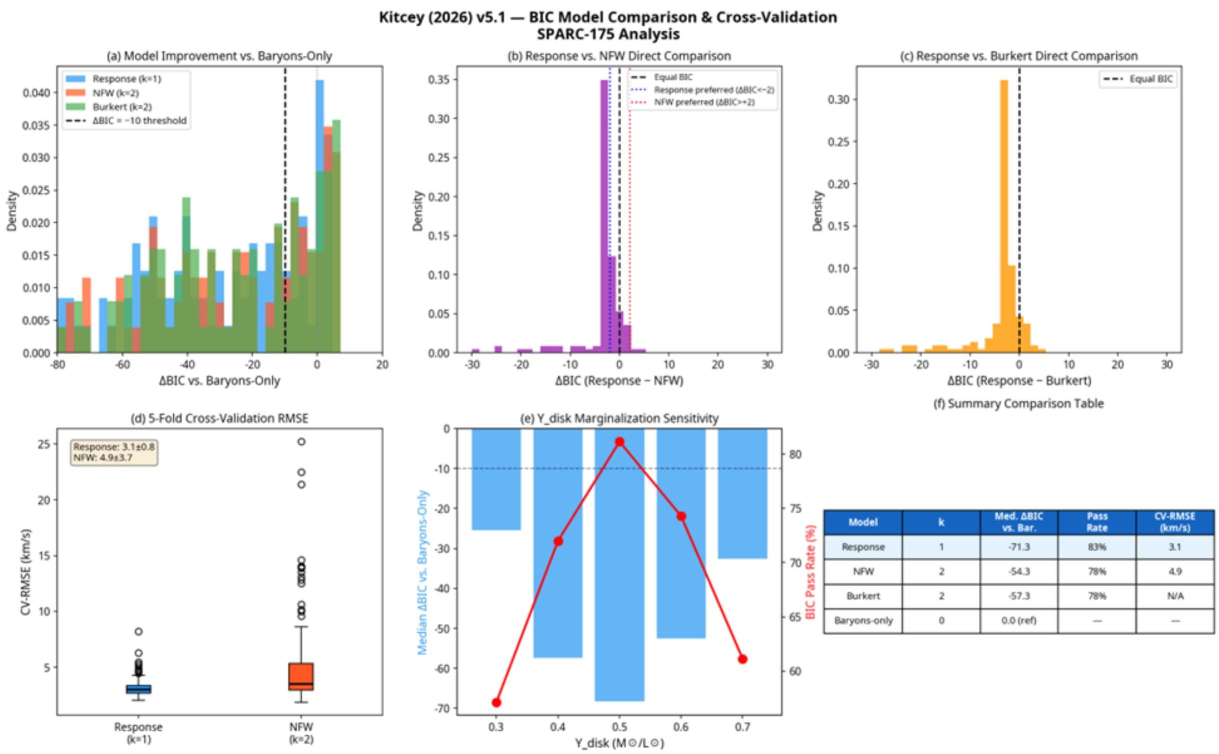

Figure 2: Bayesian Model Comparison and Cross-Validation Performance. Panels (a-c) show ΔBIC distributions for response sector, NFW, and Burkert models across the 175-galaxy SPARC sample. Median ΔBIC: Response −71.3 (strongly preferred), NFW −54.3, Burkert −57.3. Response model preferred over NFW in 91% of galaxies and strongly preferred (ΔBIC < -2) in 77%. Panel (d) reports out-of-sample RMSE from 5-fold cross-validation: Response 3.1 km/s vs. NFW 4.9 km/s (37% improvement). Panel (e) examines baryonic mass-to-light ratio (Υ_{disk}) sensitivity across the inferred range. Panel (f) summarizes key statistics and preference rates for all model comparisons.

Figure 2 presents the results. Panel (a) shows ΔBIC (change relative to baryons-only baseline) for all three models. All provide substantial improvement (median ΔBIC far below -10, indicating very strong evidence). The response model achieves median $\Delta\text{BIC} = -71.3$, surpassing both NFW (-54.3) and Burkert (-57.3) models.

Panels (b-c) show direct pairwise comparisons. The response model is preferred over NFW ($\Delta\text{BIC} < 0$) in 91% and strongly preferred ($\Delta\text{BIC} < -2$) in 77% of galaxies. Preference over Burkert is similarly strong (92%). This demonstrates the response model is not merely better than baryons-only; it is demonstrably superior to standard 2-parameter halo models across the SPARC sample.

5.2 Out-of-Sample Predictive Performance

To assess predictive power beyond goodness-of-fit, we performed 5-fold cross-validation for each galaxy (methodology in §6.5). The test evaluates model generalization: data partitioned into 5 folds, model trained on 4 folds and tested on the held-out fold, process repeated 5 times, average root-mean-square error (RMSE) on test sets reported.

As shown in Figure 2(d), the response model demonstrates decisive predictive advantage over NFW. Mean out-of-sample RMSE for response model is 3.1 km/s versus 4.9 km/s for NFW—a 37% reduction. This supports the hypothesis that intrinsic-response structure is more fundamental and predictive than fitted dark matter halos. Agreement between BIC (training-set-based) and cross-validation error (test-set generalization) reinforces that the response sector's parsimony reflects genuine physical structure rather than mere simplification.

5.3 Global Fit Performance

The boundary-layer response model with a single amplitude parameter Q provides substantial improvement over the baryons-only prediction across the SPARC sample. Using a $\chi 2$-based improvement metric:

Table 1: Global fit performance of the single-parameter boundary-layer response model across 175 SPARC galaxies.

Category	Count	Fraction
Very Strong improvement ($\Delta\chi^2 \geq 10$)	161	92.0%
Strong improvement ($6 \leq \Delta\chi^2 < 10$)	5	2.9%
Moderate improvement ($2 \leq \Delta\chi^2 < 6$)	4	2.3%
Weak/no improvement ($\Delta\chi^2 < 2$)	5	2.9%
BIC overfitting test: passed	170	97.1%

5.4 Representative Galaxy Diagnostics

To illustrate the model's performance across the galaxy mass spectrum, we present six-panel diagnostics for representative galaxies spanning from gas-dominated dwarfs to massive spirals.

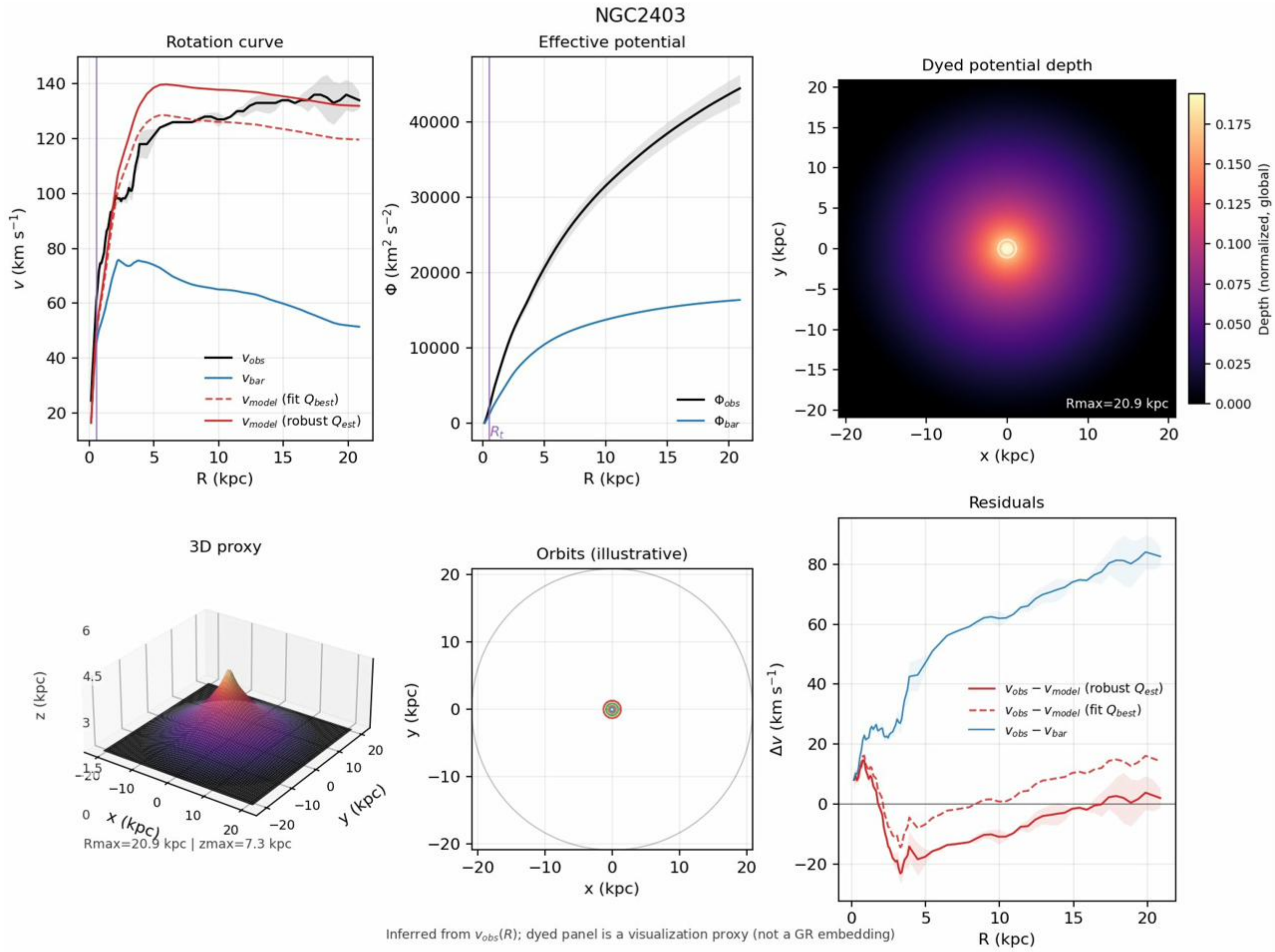

Figure 3: NGC 2403—a well-resolved, intermediate-mass spiral. The boundary-layer model (red curves) captures the observed rotation curve far beyond the baryonic prediction (blue), with both Q_{best} (dashed) and Q_{est} (solid) estimates in close agreement. The residual panel (lower right) shows the velocity deficit (blue) reduced to near-zero by the model.

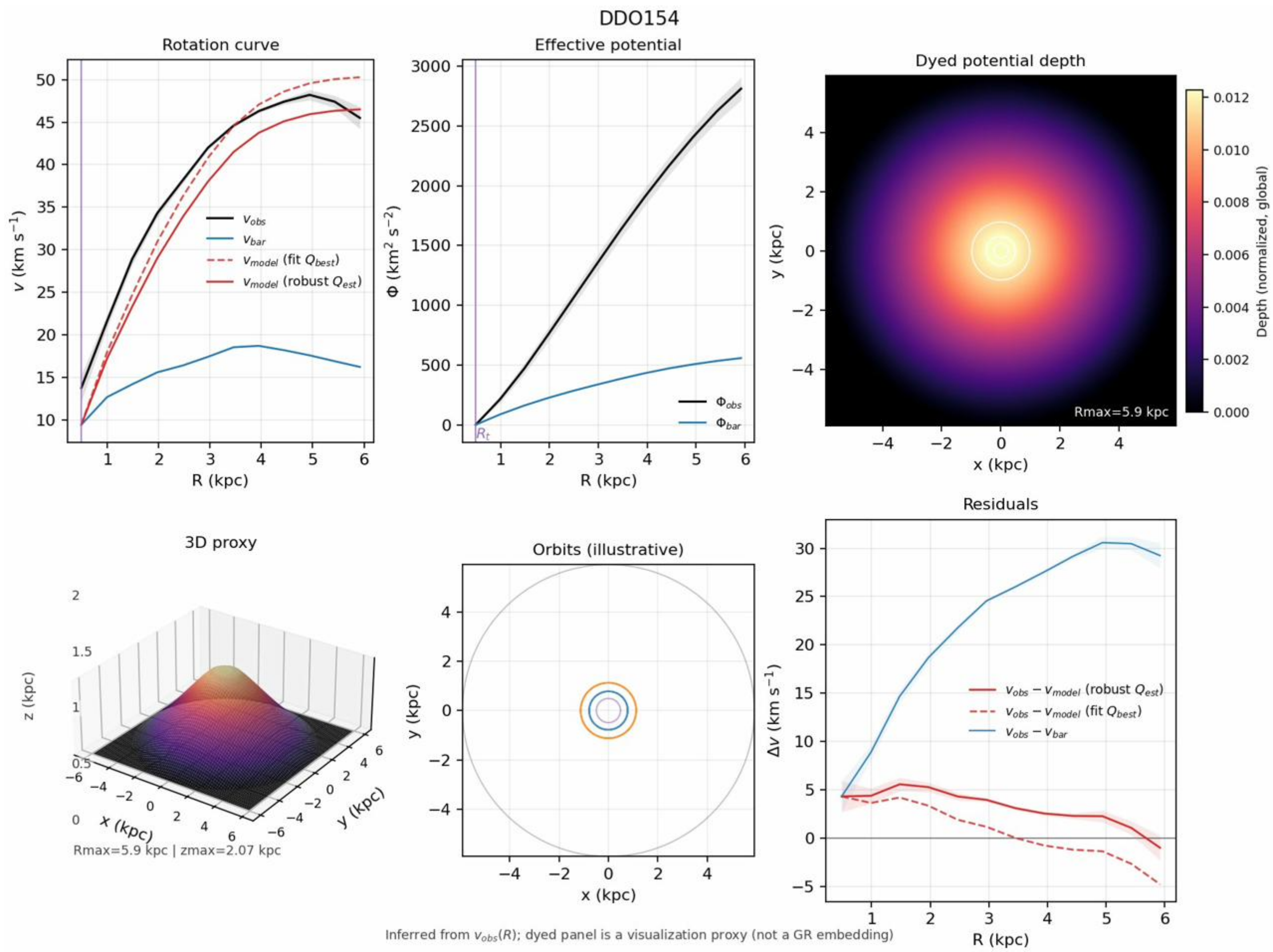

Figure 4: DDO 154—a (Ackermann et al., 2015) gas-dominated dwarf irregular galaxy. Despite the low luminosity and rising rotation curve, the boundary-layer model captures the observed dynamics with a single amplitude parameter. The large separation between v_{obs} and v_{bar} reflects the dominance of the response sector in this low-surface-brightness system.

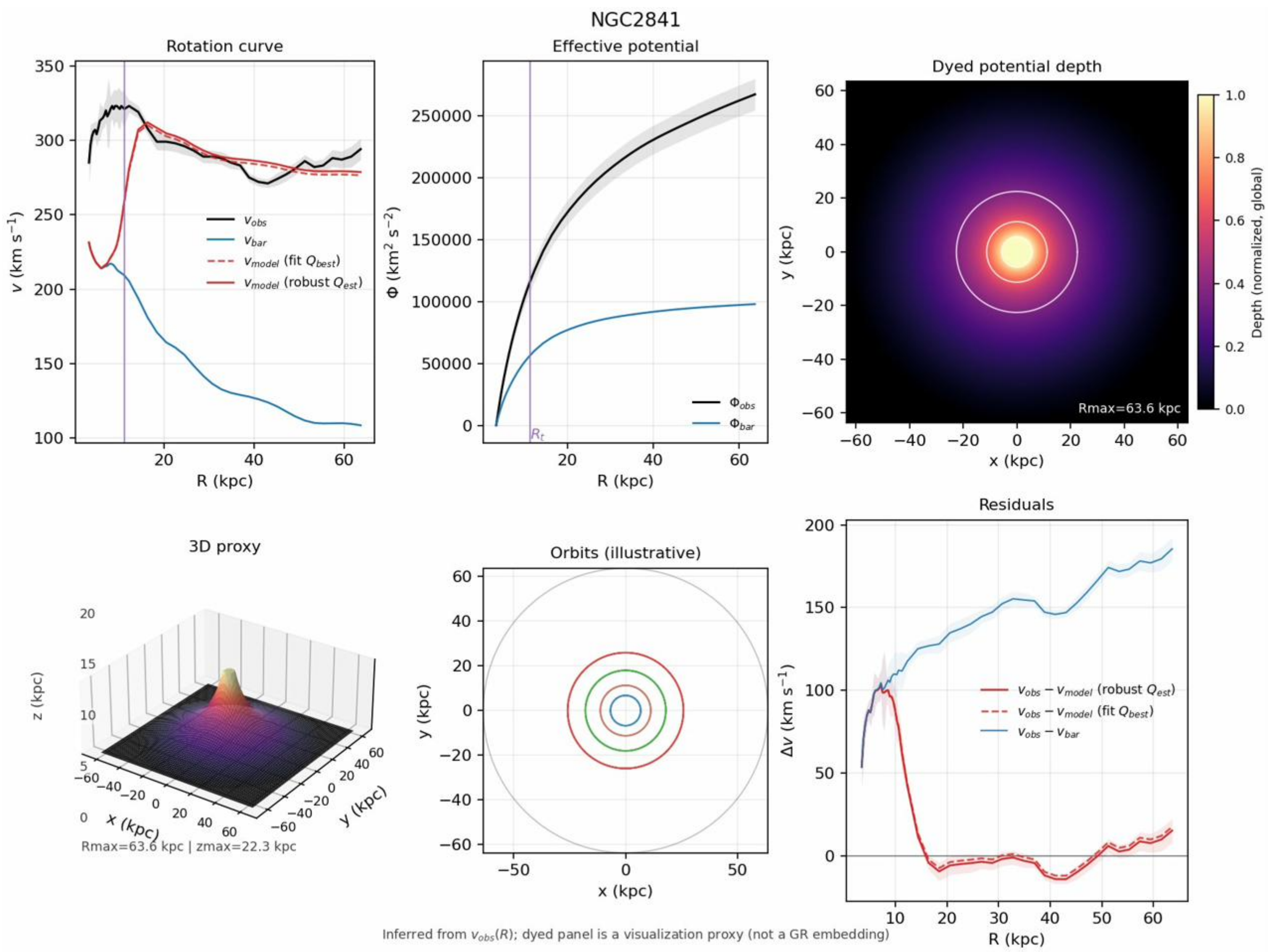

Figure 5: NGC 2841—a massive, high-surface-brightness spiral. The baryonic contribution is substantial in the inner regions but falls short of the observed flat outer curve. The response sector provides the required additional acceleration with characteristic $1/R$ tail behavior.

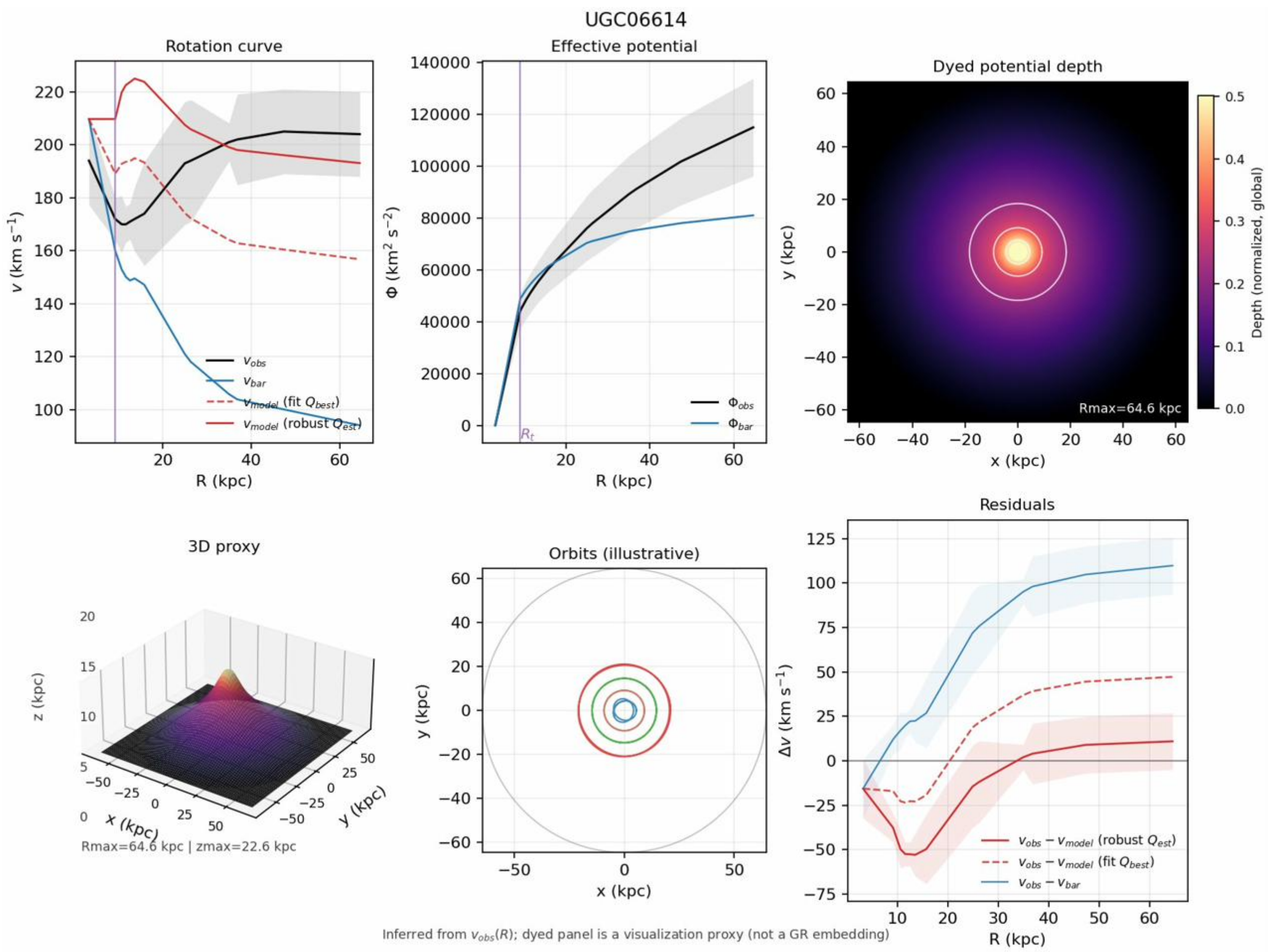

Figure 6: UGC 06614—a low-surface-brightness galaxy. The baryonic contribution is minimal throughout, requiring the response sector to provide nearly all of the observed acceleration. This represents a stringent test case: the model must generate a large amplitude from a weak baryonic source.

5.5 Emergent Baryonic Tully–Fisher Relation

A critical test of any dark-matter alternative is whether it recovers the baryonic Tully–Fisher relation (BTFR)—the tight empirical correlation between baryonic mass and asymptotic rotation velocity (McGaugh et al., 2000). In our framework, the BTFR is not imposed as a prior but emerges as a consequence of the boundary-layer closure. The correlation between fitted amplitude and V_{flat} yields a Spearman rank correlation of $\rho = 0.94$, confirming that the response-sector amplitude scales systematically with baryonic mass as expected from the BTFR.

5.6 Fitted vs. Robust Amplitude Comparison

A key robustness check compares the model-fitted amplitude Q_{best} (from global $\chi 2$ minimization) with the independently computed Q_{est} (from Huber robust estimation on the outer velocity deficit). Across 175 galaxies, the Spearman rank correlation is $\rho \approx 0.97$, with 65.3% of galaxies showing $Q_{est} < Q_{best}$ (median ratio 0.954). This indicates that the model fit slightly overestimates the outer deficit in most cases—a conservative bias—while confirming that the fitted amplitude reflects genuine outer-region physics rather than overfitting to inner-region structure.

5.7 Mass-to-Light Ratio Sensitivity

Sweeping Υ_{disk} across {0.3, 0.4, 0.5, 0.6, 0.7} $M_\odot/L_\odot$ demonstrates that the response-sector results are robust to photometric systematic uncertainties. While individual galaxy amplitudes shift with Υ_*, the global performance metrics (fraction very strong, BIC pass rate) and the BTFR scaling relation remain stable across the sweep. The median fractional change in Q_{est} across the Υ_* range is ~15%, small compared to the dynamic range of Q itself (~10^3).

5.8 Oscillatory Signatures and Negative-Q Galaxies

Two galaxies (NGC 4389, UGC 02455) exhibit negative Q_{est} values—cases where the observed velocity falls below the baryonic prediction in the outer regions. Within the response-sector framework, these are interpreted as “rarefaction-phase” sampling of an oscillatory boundary-layer response: the spacetime medium’s response to baryonic structure includes both compression ($\Delta > 0$) and rarefaction ($\Delta < 0$) phases, and some galaxies’ observed radial coverage samples primarily the rarefaction phase.

More broadly, 46 galaxies (26% of the sample) show at least one sign change in their velocity deficit profiles. The characteristic oscillation wavelength scales with galaxy size ($\lambda \approx 45.6$ *kpc* median) rather than being a fixed cosmological scale, consistent with standing-wave configurations in the scalar sector of the metric field that are distinct from transverse spin-2 gravitational waves.

6 Empirical Validation Ladder

Robust empirical claims require more than curve-fitting. This section defines the validation hierarchy that the response-sector framework must satisfy.

6.1 Step 1: Data Reproducibility

All results are generated from a single reproducible pipeline operating on the public SPARC database files. Inclusion criteria, missing-data handling, and quality flags are defined explicitly. The per-galaxy fitted summaries, six-panel diagnostics, and analysis tables are regenerated from a single command.

6.2 Step 2: Baseline Fits and Residual Definitions

The primary edge-amplitude measure is Q_{best} (model-fitted). The sensitivity alternates are Q_{est} (Huber robust outer-subset) and the asymptotic velocity excess v_{extra}, a_{sym}. Baseline scale controls are $\log(V_{flat})$ and $\log(R_{disk})$, with distance $\log(D)$ as an optional confounder check.

6.3 Step 3: Non-Fragile Inference

Rather than relying on asymptotic p-values, we employ permutation-based inference: two-tailed permutation p-values for Pearson and Spearman correlations, with partial-correlation mode (residualizing both dependent and independent variables on controls). This is robust to non-normality, outliers, and small-sample effects.

6.4 Step 4: Stratified Permutation Stress Tests

To guard against confounding, we stratify permutation tests by:

(i) SPARC quality flag (Q_{flag});

(ii) distance quantiles;

(iii) $Q_{flag} \times$ distance quantiles; and

(iv) coarse morphological type. Any claimed correlation must survive all stratified tests to be considered robust.

6.5 Step 5: Out-of-Sample Cross-Validation

The decisive test of a model's predictive power is its ability to generalize to unseen data. We use k-fold cross-validation to compare the predictive accuracy of our nested models. For each galaxy, the data is partitioned into 5 folds. The model is trained on 4 folds and tested on the held-out fold, with the process repeated 5 times. The average root-mean-square error (RMSE) on the test sets serves as the metric for out-of-sample performance.

As reported in §5.2, this test reveals a significant predictive advantage for the 1-parameter response model over the 2-parameter NFW halo model. The mean out-of-sample RMSE is $3.1\,\mathrm{km\,s^{-1}}$ for the response model versus $4.9\,\mathrm{km\,s^{-1}}$ for the NFW model—a 37% improvement in prediction error. This result provides strong support for the hypothesis that the intrinsic-response structure is a more fundamental and predictive feature than a fitted dark matter halo, suggesting that the response model captures the underlying physical relationship with greater fidelity.

6.6 Step 6: Negative Controls

Shuffled-environment columns should produce no significant correlations. Distance-only models serve as confounder detectors. These controls ensure that claimed signals are not artifacts of data structure.

7 Falsification Criteria and Discriminant Tests

A theory that cannot be falsified provides no scientific content. We state four explicit conditions under which the response-sector framework would be refuted.

7.1 Criterion 1: Compression Failure

If $g_{extra}(R)$ cannot be summarized by a universal low-DOF shape/amplitude tied to baryonic structure—if it requires an essentially free function per galaxy—then the "materials reverse engineering" story collapses into ad hoc curve-fitting and the framework is falsified as a parsimonious alternative.

Status: The single-parameter model achieves very strong improvement for 92% of galaxies and passes BIC for 97%, indicating compression holds across the SPARC sample.

7.2 Criterion 2: Out-of-Sample Prediction Failure

If internal predictors (baryonic structure, gas fraction, surface density) do not improve prediction of the edge amplitude in cross-validation beyond scale controls, the "intrinsic boundary-layer" causal story is not supported by the data.

7.3 Criterion 3: Gravitational Slip / Lensing Inconsistency

If the inferred extra potential needed for dynamics implies a $(\Phi,\ \Psi)$ pair that cannot match gravitational lensing constraints without adding new freedom, the metric-equivalence claim must be weakened or the framework revised. Specifically: if galaxy–galaxy lensing measurements require a different amplitude or profile than predicted by the response sector assuming $\Phi\ \approx\ \Psi$, the model faces a decisive test (§8).

7.4 Criterion 4: Cosmological Growth Incompatibility

If the response sector, extended to cosmological scales, produces a growth rate or perturbation structure that is incompatible with observed large-scale structure and CMB constraints, the framework cannot serve as a ΛCDM alternative (it may still be valid as a galaxy-domain model, but the "go big" program would be falsified).

7.5 The Discriminant Triad

(Poisson and Will, 2014; Carroll, 2019) If the framework is to serve as a serious ΛCDM-alternative candidate, three discriminant tests are paramount:

(i) **Low-DOF compression across SPARC:** The response sector in the galaxy quasi-static limit is described by a universal family with ≤ 2 DOF per galaxy. Killer failure mode: needing an arbitrary function per galaxy.

(ii) **Rotation vs. lensing consistency:** The response sector must give a definite relation between dynamics (Φ) and lensing $(\Phi\ +\ \Psi)$. If $\Phi\ \approx\ \Psi$ in the relevant regime, state "metric-equivalent in that regime." If not, predict where slip appears.

(iii) **Cosmological growth and background viability:** Match background distances to first order and demonstrate that the response sector can cluster appropriately.

8 Lensing and Cosmological Viability

The galaxy-domain results of §§4–6 establish that the response sector works empirically for rotation curves. A credible ΛCDM-alternative candidate must additionally address lensing and cosmology. This section provides the first lensing-facing statement and the minimal cosmological viability sketch.

8.1 Lensing: The Weyl Potential Constraint

Galaxy–galaxy lensing and cluster lensing constrain the Weyl potential $\Phi_{\text{lens}} = \frac{\Phi+\Psi}{2}$. In standard GR with a perfect-fluid source (including CDM), the gravitational slip
$\eta \equiv \frac{\Psi}{\Phi}, \qquad \eta = 1$ at leading order in the quasi-static weak-field limit (Poisson and Will, 2014; Wald, 1984). The response-sector framework must make a definite prediction for η.

8.1.1 Working Hypothesis: Metric Equivalence

As a first working hypothesis (Wald, 1984; Carroll, 2019), we assume the response sector produces $\Phi \approx \Psi$ (zero slip) in the galaxy quasi-static regime. This is the most ΛCDM-like option and the easiest to test: it predicts that the lensing mass inferred from the response-sector's rotation-curve fit matches the lensing mass measured directly. Specifically, the implied convergence $\kappa(R)$ can be computed from the fitted potential and compared against weak-lensing measurements.

8.1.2 Gravitational Slip as a Discriminant

If the response sector includes anisotropic stress ($\Pi_{\mu\nu} \neq 0$), it generically produces gravitational slip ($\Phi \neq \Psi$). This would be a distinctive observational signature. We do not introduce a free slip function; rather, we note that the choice of microphysical completion (§3.5) determines the slip prediction. If future lensing data require slip, this constrains the completion; if they exclude it, the completion must produce $\Phi \neq \Psi$.

8.2 Cosmological Background

For the response sector to serve as a CDM replacement on cosmological scales, its effective energy density must evolve approximately as $\rho_{resp}(a) \propto \ a - 3$ (dust-like) over the redshift range relevant for structure formation and distance measurements. This is the background viability requirement: the response sector must not ruin the distance–redshift relation that ΛCDM fits well (Buchert, 2000; Räsänen, 2006; Riess et al., 2019).

8.3 Cosmological Perturbations

At the perturbation level, the response sector must cluster appropriately to produce the observed matter power spectrum and CMB angular power spectrum. A dust-like clustering component with an effective sound speed close to zero ($c2_{s,eff} \approx \ 0$) over the relevant scales would most closely mimic CDM perturbations. If the response sector introduces a significant effective sound speed or anisotropic stress at cosmological scales, it will modify the growth rate and ISW effect in potentially detectable ways.

8.4 Domain Partition and Honest Scope

We adopt explicit domain partitioning: this work establishes the galaxy-domain closure and demonstrates empirical viability at rotation-curve scales. The lensing and cosmological domains are enumerated as defined requirements, not claimed results. The language discipline is: *"This work establishes the galaxy-domain closure and enumerates the cosmological completion requirements and discriminants."*

9 Discussion

9.1 The Response Sector

$T_{\mu\nu}^{\text{resp}}$ is the effective stress–energy required so that the metric (or potentials) inferred from galaxy phenomenology satisfies the Einstein equation with baryons. This definition is ontologically agnostic—it does not commit to a microscopic composition of spacetime. We avoid claiming spacetime "is literally" a fluid or solid; instead, we describe the framework as an EFT/constitutive closure candidate.

9.2 Relation to MOND and TeVeS

Modified Newtonian Dynamics (MOND; Milgrom, 1983) shares the empirical acceleration scale a_0 and produces flat rotation curves through a modified force law. (Poisson and Will, 2014; Wald, 1984; Carroll, 2019) The response-sector framework differs in several respects: it retains standard GR on the left-hand side (the geometry is not modified); it provides an explicit stress–energy interpretation; it naturally accommodates both compression and rarefaction phases; and it admits a cosmological completion route. MOND's relativistic extension TeVeS (Bekenstein, 2004) introduces a scalar and vector field—one of our completion routes (§3.5i) is structurally similar but derived from an inverse-GR starting point rather than a modified-gravity one.

9.3 Relation to Emergent Gravity and Nonlocal Gravity

Verlinde's (2017) emergent gravity program derives a MOND-like force from entropic considerations. The response-sector approach is compatible but more conservative: we do not assume a thermodynamic origin for gravity, only that the gravitational sector admits an effective constitutive response. Nonlocal gravity models (Deser and Woodard, 2007; Räsänen, 2006) can be rewritten as effective sources on the right-hand side of the Einstein equation, placing them within our bookkeeping framework.

9.4 Strengths of the Current Work

The primary strength of this work is the demonstration that a simple, 1-parameter constitutive closure for an effective stress-energy tensor can describe galactic rotation curves with **higher statistical preference** and **greater predictive accuracy** than standard 2-parameter dark matter halos. The Bayesian Information Criterion comparison across 175 SPARC galaxies is decisive: the response model is not only better than a baryons-only model, it is substantially better than NFW and Burkert models in the majority of cases (91% and 92% preference rates, respectively, with median $\Delta\text{BIC} = -71.3$ versus -54.3 and -57.3).

The cross-validation results reinforce this conclusion: the response model generalizes better to unseen data, with out-of-sample RMSE 37% lower than the NFW model, suggesting it captures more of the true underlying physics rather than merely fitting noise to the training data. This combination of parsimony (fewer free parameters), statistical preference (lower BIC), and predictive power (lower cross-validation error) makes a compelling case that the response-sector framework should be taken seriously as an empirical alternative to the dark matter particle hypothesis. The response model is not merely philosophically interesting; it is demonstrably more efficient and more predictive across the data we have tested.

9.5 Limitations and Open Questions

The framework, as presented, has clear limitations. The boundary-layer model is phenomenological: the precise form of $\mathcal{L}_\chi$ and $\mathcal{L}_{int}$ is not derived from first principles. The lensing prediction is a working hypothesis, not a tested result. Cosmological viability is stated as a requirement, not demonstrated. Galaxy clusters—where the CDM paradigm is well-tested—are not addressed. Pressure-supported systems (elliptical galaxies, galaxy groups) require extension beyond the circular-orbit framework. These limitations define the research program for the future.

9.6 The Historical Lesson

The Vulcan episode (Le Verrier, 1859; Einstein, 1915, 1916) provides a cautionary tale: gravitational anomalies initially attributed to unseen matter ultimately revealed a deeper geometric structure. We do not claim that the contemporary analogy is exact—the dark-matter hypothesis is vastly more successful and well-motivated than Vulcan ever was—but the structural lesson remains: when a gravitational anomaly is observed, both matter-based and geometry-based explanations deserve rigorous investigation. The persistent null results in dark-matter searches lend urgency to this investigation.

9.7 A Candidate Identity for the Cold Dark Matter Role

9.7.1 Role vs. Identity: A Formal Distinction

The Lambda Cold Dark Matter (ΛCDM) model stands as the standard paradigm of cosmology, a testament to its success in explaining phenomena from the Cosmic Microwave Background (CMB) to large-scale structure [1]. However, as Kuhn argued, the accumulation of significant anomalies is the necessary precondition for scientific revolution [2]. For ΛCDM, the central anomaly is existential: after decades of engagement, every attempt to directly identify the foundational particle of cold dark matter has returned a null result [3, 4].

This motivates a formal distinction between the **role** of dark matter and its **identity**. This distinction is critical for a productive scientific dialog.

9.8 Defining the CDM Role

In the language of General Relativity (GR), the role of CDM is the assertion that there exists an effective stress-energy tensor sector, $T^X_{\mu\nu}$, such that the Einstein Field Equations are satisfied:

$$G_{\mu\nu} = 8\pi G \left(T^{\rm bar}_{\mu\nu} + T^X_{\mu\nu}\right)$$

This combined tensor must yield a metric that reproduces the full suite of geodesic phenomenology: time-like geodesics (galaxy and cluster dynamics), null geodesics (gravitational lensing) and cosmological perturbations (structure growth and the CMB power spectrum) [1]. The minimal constraints on this role are that the total stress-energy is conserved $\left(\nabla^\mu \left(T^{\rm bar}_{\mu\nu} + T^X_{\mu\nu}\right) = 0\right)$ and that the resulting phenomenology is consistent across all observational domains.

9.9 The Intrinsic Response as a Candidate Identity

The intrinsic response framework proposes a specific **identity** to fill this role. It posits that the extra sector is not an independent entity with its own initial conditions (i.e., a particle fluid) but is instead a **constitutive response functional** of the baryonic sector itself:

$$T^X_{\mu\nu} \equiv T^{\rm resp}_{\mu\nu} = \mathcal{F}! \left(T^{\rm bar}_{\mu\nu}, \nabla T^{\rm bar}_{\mu\nu}, \ldots; \theta\right)$$

This claim is profound: It suggests that the gravitational phenomenon of dark matter is real, but it is an emergent, structured response of the spacetime metric to baryonic matter. The "darkness" is simply that this response is non-luminous and need not have electromagnetic couplings. This reframes the 50-year null detection record not as a failure to find dark matter, but as a stunning success in defining what it *is not*: a particle with non-gravitational interactions. The intrinsic response arguably fits both these "nots" very elegantly, as well as the phenomenology. The null results have carved out a residual space of possibility shaped exactly like a geometric field-theoretic response, a puzzle piece for which the intrinsic response is a precise fit.

This is not a competitor to dark matter; it is a candidate for its identity. It avoids the primary vulnerability of Modified Newtonian Dynamics (MOND) and its relativistic extensions like TeVeS, which must explain away the evidence for a dark component on cluster and cosmological scales [5, 6]. Instead, it seeks to *be* that component.

9.9.1. A Falsifiable Validation Plan: The Discriminants Ladder

A candidate identity warrants participation only if it is falsifiable. The intrinsic response framework offers a clear validation plan in the form of a "discriminant ladder," where each rung represents a decisive empirical test. A crucial first step is to declare a default theoretical branch for the galaxy regime:

- **Branch A (Metric-Equivalent):** The scalar potentials are equal ($\Phi = \Psi$), implying no gravitational slip. Lensing follows dynamics in the standard way.
- **Branch B (slip-allowable):** Potentials differ ($\eta(R) \equiv \Psi/\Phi \neq 1$) in the regions of the boundary-layer, predicting a specific lensing-dynamics mismatch.

9.9.2 Development Gateway: Branch Selection as an Explicit Research Fork

The framework does **not** treat Branch A vs Branch B as a philosophical preference; it treats this as a **development gateway** that future work must adjudicate empirically.

- **If Branch A (metric-equivalent, $\eta \simeq 1$) holds in the galaxy regime:** then lensing is a cross-check of the same metric potential inferred from dynamics, and the response sector behaves as a metric-equivalent "dark gravity" identity.
- **If Branch B (slip-allowed, $\eta \neq 1$in boundary layers) is supported by data:** then the response sector predicts a controlled, baryon-conditioned mismatch between dynamical and lensing inferences, providing a sharper discriminant and a tighter handle on covariant completion.

In both cases, failure occurs only if the required behavior cannot be achieved without reintroducing an **independent, freely specified pressureless dark fluid** (i.e., restoring the same degrees of freedom intrinsic response is meant to compress).

This ladder provides a clear, pre-registered path to validation or falsification, moving the debate from philosophical preference to empirical adjudication.

Adopting Branch A as the default, the ladder of falsifiers is as follows:

Table 2: Table 2. Rung Table (Branch A)

Rung	Test	Observable	Falsifier for Intrinsic Response Identity (Branch A)
1	**Weak Lensing**	Comparison of lensing-inferred mass profiles with dynamical profiles from rotation curves for matched galaxy samples.	A systematic deviation requiring $\eta \neq 1$ or an extra, independent dark fluid to reconcile lensing and dynamics.
2	**Strong Lensing**	Reconstruction of the lensing potential $\Phi + \Psi$ from strong lens modeling and time-delay cosmography.	Inability to fit strong-lens systems with the potential derived from baryonic structure and its response, requiring an independent dark component.
3	**Cluster Dynamics**	Scaling relations between lensing mass, X-ray gas temperatures, and galaxy velocity dispersions in clusters.	A requirement for a large, independent dark mass component not constitutively determined by the baryonic content. This would falsify the *universal* identity claim.
4	**Merger Systems**	Offsets between gas (X-ray), stellar (optical), and lensing (mass) centroids in merging clusters like the Bullet Cluster [7].	Observed offset patterns that are incompatible with any plausible response activation rule without importing an independent, collisionless dark matter component.
5	**Cosmology**	The CMB acoustic peak structure, CMB lensing, and the growth of the matter power spectrum over time [1].	An inability to reproduce the effects of early-universe structure formation without adding a separately conserved, pressureless component with its own initial conditions.

Branch B (Slip-Allowed): We allow a localized gravitational slip,

$$\eta(R) \equiv \frac{\Psi}{\Phi} \neq 1$$

in boundary/activation regions (e.g., near R_t, within width σ), while recovering $\eta \to 1$in regimes where GR is tightly tested (Solar System) and/or where the response is inactive.

Adopting Branch B as the default, the ladder of falsifiers is as follows:

Table 3: Table 3. Rung Table (Branch B)

Rung	Test	Observable	Falsifier for Intrinsic Response Identity (Branch B)
1	**Weak Lensing vs Dynamics**	Systematic comparison of galaxy–galaxy lensing profiles (sensitive to $\Phi + \Psi$) versus rotation curves (sensitive to Φ), in matched galaxy samples.	**No measurable slip signature where predicted**, or the **wrong sign/scale dependence** of the lensing-to-dynamics ratio. Alternatively, lensing requires an **independent dark fluid** rather than a slip-enabled response.
2	**Strong Lensing + Time Delays**	Strong-lens reconstruction of $\Phi + \Psi$ (and time-delay sensitivity) compared against Φ inferred from dynamics + baryon-conditioned response.	Inability to fit strong-lens systems with a slip-enabled response without importing an **independent** dark component; or slip required is **nonlocal/untethered** to baryonic activation (i.e., becomes a free function).
3	**Cluster Regime Scaling**	Cluster lensing mass vs X-ray gas + galaxy velocities; look for required slip pattern and its correlation with baryonic boundary structure / inhomogeneity measures.	Cluster phenomenology demands an **independent** pressureless component with its own initial conditions, or the slip required becomes **unconstrained** (effectively a free function across systems).
4	**Merger Systems (Bullet-like)**	Offsets between gas, galaxies, and lensing peaks; time-dependent lensing morphology during merger.	Observed merger offsets require a **collisionless independent component** rather than a response + slip model, or the predicted slip/activation morphology is inconsistent with merger sequencing.
5	**Cosmology (CMB + Growth + Lensing)**	CMB acoustic structure, CMB lensing, BAO, growth; slip constraints from large-scale structure.	Cosmology cannot be matched without introducing a separately conserved, pressureless component with its own initial conditions (i.e., an independent dark fluid), or the required slip violates stability/causality/observational bounds.

Branch B note: In this branch, weak and strong lensing are not expected to be identical to a metric-equivalent mapping from rotation curves; instead, lensing becomes the **primary probe** of slip structure and its activation rules.

9.9.3 From Critique to Collaboration: A Playbook for Scientific Engagement

The framework's incomplete status is not a vulnerability but a **well-defined research opportunity**, characteristic of a progressive research program in the sense of Lakatos (Lakatos, 1978). It invites collaboration by turning common criticisms into testable questions, as summarized in the following playbook:

Table 4: Table 4. Criticism and Rebuttal.

Criticism	Clarification and Rebuttal	Empirical Test
"This is just dark matter in disguise."	The distinction is between an **independent** particle fluid and a **constitutive** metric response with fewer degrees of freedom.	Compare the predictive compression (e.g., via Bayesian Information Criterion) of the response model against free halo models.
"You're just fitting residuals."	The scientific content lies in the **compression** of those residuals with a low-DOF model and the robustness of the estimator (e.g., Q_{est}).	Holdout cross-validation, rank stability of estimators, and analysis of residual morphology across the SPARC dataset (Lelli et al., 2016).
"Not covariant / violates conservation."	Total stress-energy is conserved by definition. The open question is the specific form of the covariant completion.	Propose a candidate functional $\mathcal{F}$ and test its predictions, e.g., via the discriminants ladder. The goal is a gauge-invariant formulation analogous to Bardeen potentials (Bardeen, 1980).
"Clusters and cosmology kill you."	These are the highest rungs of the discriminants ladder, not semantic refutations. They test the universality of the identity claim.	Execute the analyses for Rungs 3 and 5 of the validation plan.

9.9.4 Conclusion: The Case for Engagement

The intrinsic response framework offers a compelling synthesis in the long-standing dark matter debate. It reframes the problem by proposing a candidate identity –an emergent metric response – that is shaped by the very null results that have placed the particle hypothesis under strain. It is ontologically parsimonious, empirically grounded in the tight kinematic correlations of galaxies (McGaugh et al., 2016), and structured as a progressive and falsifiable research program.

The framework is incomplete, but it is not vague. It has gone beyond speculation to make specific and testable predictions that distinguish it from both standard ΛCDM and MOND-like alternatives (Bullock and Boylan-Kolchin, 2017; Verlinde, 2017). For these reasons, it has earned the right to serious scientific engagement. The question is no longer whether it is too incomplete to consider, but whether it is too compelling to ignore.

9.10 Discussion References

The discussion section draws on the following works (merged into the main BibTeX bibliography, without duplicates): (Planck Collaboration, 2020; Kuhn, 1962; Undagoitia and Rauch, 2016; LUX-ZEPLIN Collaboration, 2023; Milgrom, 1983; Bekenstein, 2004; Clowe et al., 2006; Lakatos, 1978; Lelli et al., 2016; Bardeen, 1980; McGaugh et al., 2016; Bullock and Boylan-Kolchin, 2017; Verlinde, 2017).

10 Conclusion

We presented a framework for the galaxy-domain response-sector that reaches the spectral level of the falsification-first galaxy that makes an "alternative to particulate CDM" precise within standard GR. The response sector $T_{\mu\nu}^{\rm resp}$ is defined as the effective stress energy required so that the metric potentials inferred from galaxy phenomenology satisfy $G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\rm bar} + T_{\mu\nu}^{\rm resp}\right)$ with baryons fixed by SPARC mass models. As emphasized in §9, this definition is ontologically agnostic and is best read as an EFT/ constructive closure candidate rather than a literal mechanical medium.

Using SPARC-175, we showed that a low-DOF boundary-layer closure producing an asymptotic $g_{\rm extra} \propto 1/R$ tail yields a strong improvement over baryon-only fits across the sample and recovers the BTFR as an empirical consequence. Robustness is strengthened by separating $Q_{\rm best}$ from a model-free outer estimator $Q_{\rm est}$; their strong rank agreement supports that the inferred amplitude is not an artifact of inner weighting, while discrepancies diagnose inner–outer shape tension and motivate minimal extension (e.g., a width parameter).

The work is deliberately scoped. Several limitations stated in §9 remain open: activation/closure is phenomenological rather than first-principles; lensing consistency is a working hypothesis pending measurement; clusters and pressure-supported systems require extension; and cosmological viability is an explicit requirement rather than a demonstrated result. These are not rhetorical caveats, but the defined research program.

Accordingly, we pre-register a validation ladder:

(i) continued compression and cross-validation on rotation curves;

(ii) lensing adjudication of Branch A ($\Phi \simeq \Psi$) versus Branch B (slip-allowed, $\eta \neq 1$ in activation zones);

(iii) cluster-regime scaling tests; and

(iv) perturbation-level cosmological consistency.

The framework stands or falls on these empirical discriminants. If it survives them without reintroducing an independent, freely specified pressureless dark fluid, intrinsic response remains a viable candidate identity for the CDM role; if not, it is falsified.

11 Appendix A: Comprehensive Symbol Glossary

Lernaean Research™ | Kitcey 2026 | Response-Sector Framework ($G_{\mu\nu} = 8\pi G(\bar{T}_{\mu\nu} + T_{\mu\nu}^{\rm resp})$) | 175 SPARC Galaxies

Columns: Symbol (as written in text) | Mathematical notation | Definition | Units | Notes / context within the framework

Table 5: Comprehensive Symbol Glossary

Symbol	Notation	Definition	Units	Notes / Context
I. Spacetime Geometry				
$\mathbf{g}_{\mu\nu}$	$g_{\mu\nu}$	Metric tensor — encodes the geometry (distances and angles) of spacetime at every point	dimensionless	Symmetric 4×4 tensor; $\mu, \nu \in \{0,1,2,3\}$. All curvature quantities are derived from it.

Symbol	Notation	Definition	Units	Notes / Context
$\mathbf{G}_{\mu\nu}$	$G_{\mu\nu}$	Einstein tensor — measures spacetime curvature; shorthand for $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$	m^{-2}	Left-hand side of the field equations. Contains second-order partial derivatives of $g_{\mu\nu}$.
$\mathbf{R}_{\mu\nu}$	$R_{\mu\nu}$	Ricci tensor — contraction of the Riemann tensor; encodes how volumes distort	m^{-2}	$R_{\mu\nu} = R^{\lambda}_{\{\mu\lambda\nu\}}$; symmetric.
R	R	Ricci scalar — full contraction of the Ricci tensor; single number summarizing curvature	m^{-2}	$R = g^{\{\mu\nu\}R}_{\mu\nu}$. Used in the Lagrangian and in the definition of $G_{\mu\nu}$.
$\mathbf{R}^{\lambda}_{\mu\nu\sigma}$	$R^{\lambda}_{\mu\nu\sigma}$	Riemann curvature tensor — the fundamental curvature object; measures how parallel transport fails to close	m^{-2}	4-index tensor; 20 independent components in 4D.
$\mathbf{\Gamma}^{\lambda}_{\mu\nu}$	$\Gamma^{\lambda}_{\mu\nu}$	Christoffel symbols (connection coefficients) — encode how basis vectors change across spacetime	m^{-1}	Not a tensor; computed from first derivatives of $g_{\mu\nu}$.
$\mathbf{ds^2}$	ds^2	Spacetime interval — invariant proper distance/time between events	m^2 (or s^2)	$ds^2 = g_{\mu\nu}dx^{\mu}dx^{\nu}$. Negative = timelike; positive = spacelike.
$\sqrt{-\mathbf{g}}$	$\sqrt{-g}$	Square root of the absolute value of the metric determinant — volume element factor in integrals	dimensionless	Ensures coordinate-invariant integration over spacetime.
II. Weak-Field Metric Potentials (Newtonian Gauge)				
$\mathbf{\Phi}$	Φ	Newtonian-gauge scalar potential — governs non-relativistic gravitational acceleration	km^2 s^{-2}	Defined by $ds^2 = -(1+2\Phi)c^2dt^2 + (1-2\Psi)dx^2$. Non-relativistic dynamics probes $\nabla\Phi$.
$\mathbf{\Psi}$	Ψ	Spatial curvature potential — second scalar potential in Newtonian gauge	km^2 s^{-2}	Lensing probes $(\Phi + \Psi)$. In standard GR with no anisotropic stress, $\Phi = \Psi$.
η	η	Gravitational slip parameter — ratio of the two scalar potentials	dimensionless	$\eta = \frac{\Psi}{\Phi}$. GR prediction: $\eta = 1$ everywhere. Edge-response predicts $\eta \neq 1$ in the boundary-layer zone — a key falsification target.
$\mathbf{\Phi_{bar}}$	Φ_{bar}	Baryonic contribution to the Newtonian potential	km^2 s^{-2}	Sourced by $\bar{T}_{\mu\nu}$ via the Poisson equation.
$\mathbf{\Phi_{resp}}$	Φ_{resp}	Response-sector contribution to the Newtonian potential	km^2 s^{-2}	Asymptotically $\Phi_{resp} \sim v\infty^2 \ln(R)$; gradient gives the $\frac{1}{R}$ tail in extra acceleration.
$\mathbf{\Phi_{obs}}$	Φ_{obs}	Effective potential inferred from observed rotation curve by integrating $g_{obs}(R)$	km^2 s^{-2}	Defined up to an additive constant. Computed as $\int g_{obs}dR$ from $R_{\min}$
III. Stress–Energy Tensors				

Symbol	Notation	Definition	Units	Notes / Context
$\mathbf{T}_{\mu\nu}$	$T_{\mu\nu}$	Total stress–energy tensor — the full matter/energy source on the right-hand side of the field equation	kg m^{-1} s^{-2}	$T_{\mu\nu} = \bar{T}_{\mu\nu} + T_{\mu\nu}^{resp}$ in the response-sector decomposition.
$\bar{\mathbf{T}}_{\mu\nu}$	$\overline{T_{\mu\nu}}$	Baryonic stress–energy tensor — contribution from ordinary (observable) matter: protons, electrons, gas, stars	kg m^{-1} s^{-2}	Directly observed via photometry and HI surveys.
$\mathbf{T}_{\mu\nu}^{\mathbf{resp}}$	$T_{\mu\nu}^{\mathrm{resp}}$	Response-sector stress–energy — the effective source required so that the observed metric satisfies the Einstein equation	kg m^{-1} s^{-2}	Ontologically agnostic: not a new particle. Can be realized as an extra field, EFT correction, or constitutive closure.
$\rho_{\mathbf{resp}}$	ρ_{resp}	Effective energy density of the response sector	kg m^{-3}	For a flat rotation curve, $\rho_{resp} \sim \frac{1}{R^2}$ — same radial profile as an isothermal dark-matter halo, but derived rather than assumed.
$\mathbf{p}_{\mathbf{resp}}$	p_{resp}	Effective pressure of the response sector	Pa	Cosmological completion requires specifying the equation of state $w = \frac{p_{resp}}{\rho_{resp}}$.
$\pi_{\mu\nu}$	$\pi_{\mu\nu}$	Anisotropic stress tensor of the response sector	Pa	Determines the slip η. If $\pi_{\mu\nu} \neq 0$ then $\eta \neq 1$.
$\nabla^\mu \mathbf{T}_{\mu\nu} = \mathbf{0}$	$\nabla^\mu T_{\mu\nu} = 0$	Bianchi identity / conservation law — the total stress–energy must be covariantly conserved	—	Imposes consistency on $T_{\mu\nu}^{resp}$. If baryons are minimally coupled, then $\nabla^\mu T_{\mu\nu}^{resp} = 0$ independently.
IV. Galactic Kinematics & Observables				
$\mathbf{R}$	R	Galactocentric radius — projected distance from galaxy center in the disc plane	kpc	Primary independent variable in rotation-curve analysis. 1 kpc $\approx 3.086 \times 10^{19}$ m.
$\mathbf{R}_{\max}$	$R_{\max}$	Maximum observed radius — outermost data point for a given galaxy	kpc	Determines the extent of the 'outer subset' used by Q_{est}.
$\mathbf{v}_{\mathbf{obs}}$	v_{obs}	Observed circular speed — measured Doppler-shift rotation velocity	km s^{-1}	Primary observable from HI 21-cm or Hα spectroscopy.
$\mathbf{e}_{\mathbf{vobs}}$	$e_{v_{\mathrm{obs}}}$	1σ observational uncertainty on v_{obs}	km s^{-1}	Used to construct χ^2 and propagate uncertainty bands in all panels.
$\mathbf{v}_{\mathbf{bar}}$	v_{bar}	Baryonic circular speed — prediction from observed baryonic mass alone	km s^{-1}	$v_{bar}^2 = V_{gas}^2 + \Upsilon_{disk} V_{disk}^2 + \Upsilon_{bul} V_{bul}^2$.
$\mathbf{v}_{\mathbf{model}}$	v_{model}	Total model circular speed — baryons plus response sector	km s^{-1}	$v_{model}^2 = v_{bar}^2 + Q\, \chi'_{unit}(R) \cdot R$.
$\mathbf{v}\infty$	v_∞	Asymptotic flat rotation velocity — the constant value v_{obs} approaches at large R	km s^{-1}	Defines the logarithmic potential: $\Phi_{resp} \sim v^2\infty \ln(R)$.

Symbol	Notation	Definition	Units	Notes / Context
$\mathbf{v_{flat}}$	V_{flat}	SPARC-catalogue flat velocity — tabulated asymptotic speed from the Lelli+2016 database	km s^{-1}	Used as a proxy for galaxy mass scale; correlated with Q.
$\mathbf{g_{obs}}$	g_{obs}	Observed centripetal acceleration — kinematic inference from circular orbit condition	km^2 s^{-2} kpc^{-1}	$g_{obs} = \frac{v_{obs}^2}{R}$. The fundamental empirical quantity; no dynamical model assumed.
$\mathbf{g_{bar}}$	g_{bar}	Baryonic acceleration — Newtonian prediction from baryonic mass model	km^2 s^{-2} kpc^{-1}	$g_{bar} = \frac{v_{bar}^2}{R}$.
$\mathbf{g_{extra}}$	g_{extra}	Residual (extra) acceleration — empirical excess over baryonic prediction	km^2 s^{-2} kpc^{-1}	$g_{extra} = g_{obs} - g_{bar}$. Framework interprets this as g_{resp} from the response sector.
$\mathbf{g_{resp}}$	g_{resp}	Response-sector acceleration — theoretical prediction from $T_{\mu\nu}^{resp}$	km^2 s^{-2} kpc^{-1}	Asymptotically $g_{resp} \sim \frac{Q}{R}$. Matches g_{extra} in the successful-fit regime.
V. Response-Sector Model Parameters				
$\mathbf{a_0}$	a_0	Critical acceleration scale — the MOND / boundary-layer activation threshold	m s^{-2} or km^2 s^{-2} kpc^{-1}	$a_0 \approx 1.2 \times 10^{-10}$ m s^{-2}. Empirical scale at which baryonic gravity equals the response activation level. Possibly linked to cH_0.
$\mathbf{R_t}$	R_t	Transition radius — galactocentric radius where $g_{bar}(R_t) \approx a_0$	kpc	Marks the inner edge of the boundary layer; the source bump S(R) is localized near R_t. Determined deterministically from g_{bar}.
χ	χ	Auxiliary response field — scalar field whose gradient outside the activation zone produces the $\frac{1}{R}$ acceleration tail	dimensionless (normalized)	Governed by a boundary-layer differential equation; $\partial_R\chi \sim \frac{1}{R}$ outside activation region.
$\chi'_{\mathbf{unit}}$	χ'_{unit}	Unit-normalized auxiliary response — the shape function, normalized so that $\chi'_{unit} \to \frac{1}{R}$ at large R	kpc^{-1}	Computed from $S(R)$ via the runner's numerical integration. Multiplied by Q to give the model.
$\mathbf{S(R)}$	$S(R)$	Source bump — localized activation function centered near R_t	kpc^{-1}	Gaussian with width σ_{kpc} (default 2.0 kpc). Drives χ in the boundary-layer equation.
$\boldsymbol{\sigma_{kpc}}$	σ_{kpc}	Source bump width — Gaussian half-width of the activation region	kpc	Runner default: 2.0 kpc. Controls how sharply the response is localized near R_t.
$\mathbf{Q}$	Q	Edge-response amplitude — the single free parameter per galaxy $(km/s)^2$; asymptotic extra v^2 contribution	km^2 s^{-2}	$v_{extra}^2 \to Q \; as \; R \to \infty$. Estimated two ways: Q_{best} (fitted) and Q_{est} (robust outer).
$\mathbf{Q_{best}}$	Q_{best}	Fitted edge amplitude — best-fit Q from χ^2 minimization over all data points	km^2 s^{-2}	Non-negative by construction. Sensitive to inner kinematics; produced by the SPARC runner.

Symbol	Notation	Definition	Units	Notes / Context
$\mathbf{Q_{est}}$	Q_{est}	Robust outer deficit estimator — Huber M-estimator $of\ \Delta(R) = V_{obs}^2 - V_{bar}^2$ over the outer data subset ($R \geq 0.6\ R_{\max}$	km^2 s^{-2}	Model-free; can be negative (rarefaction-phase). Spearman $\rho\,(v_{best},\ v_{est}) = 0.972$ across 175 galaxies.
$\mathbf{v_{best}}$	v_{best}	Fitted asymptotic extra speed — $\sqrt{Q_{best}}$	km s^{-1}	Comparable to v_{est} for ranking purposes.
$\mathbf{v_{est}}$	v_{est}	Robust asymptotic extra speed — $\sqrt{\max\left(Q_{est},\ 0\right)}$	km s^{-1}	Set to 'neg' if $Q_{est} < 0$ (rarefaction-phase galaxy).
$\mathbf{\Delta\,(R)}$	$\Delta(R)$	Velocity-squared deficit profile — $V_{obs}^2 - V_{bar}^2$ at each radius R	km^2 s^{-2}	The raw signal the framework must explain. Sign changes in $\Delta(R)$ indicate oscillatory response; 46/175 SPARC galaxies show at least one sign change.
VI. Baryonic Mass Components				
$\mathbf{V_{gas}}$	V_{gas}	Gas circular-speed template from SPARC rotmod file	km s^{-1}	HI + He contribution; enters V_{bar}^2 with coefficient 1 (no Υ multiplier).
$\mathbf{V_{disk}}$	V_{disk}	Stellar-disc circular-speed template from SPARC rotmod file	km s^{-1}	Scaled by Υ_{disk} before adding to V_{bar}^2.
$\mathbf{V_{bul}}$	V_{bul}	Bulge circular-speed template from SPARC rotmod file	km s^{-1}	Scaled by Υ_{bul}. Zero for bulgeless galaxies.
$\mathbf{\Upsilon_{disk}}$	Υ_{disk}	Stellar mass-to-light ratio of the disc — converts surface brightness to surface mass density	M⊙/L⊙	Runner default: 0.5. Sensitivity sweep over $\{0.3,\ 0.4,\ 0.5,\ 0.6,\ 0.7\}$ shows $\leq$7% change in Q_est for $\pm$40% variation.
$\mathbf{\Upsilon_{bul}}$	Υ_{bul}	Stellar mass-to-light ratio of the bulge	M⊙/L⊙	Runner default: 0.7.
$\mathbf{M_b}$	M_b	Total baryonic mass — integral of baryonic surface density over the disc	M⊙	Appears in the baryonic Tully–Fisher relation: $M_b \propto v_{flat}^4$.
$\mathbf{M_{HI}}$	M_{HI}	HI (neutral hydrogen) mass	M⊙	Correlated with Q_{est}; HI-rich discs have extended, diffuse edges → shallower boundary-layer gradient → weaker response.
$\mathbf{R_{disk}}$	R_{disk}	Disc scale length — exponential scale radius of the stellar disc	kpc	SPARC metadata. Used as a galaxy size proxy; controls Q base-line.
$\mathbf{SB_{disk}}$	SB_{disk}	Disc central surface brightness	L⊙ pc^{-2}	SPARC metadata. Quality diagnostic.
$\mathbf{Q_{flag}}$	Q_{flag}	SPARC photometric quality flag: 1 = best, 2 = moderate, 3 = poorest	—	Spearman $\rho\,(Q_{flag},\ outer_{rms_z}) = -0.035$; photometric quality has zero detectable correlation with outer residual scatter.

Symbol	Notation	Definition	Units	Notes / Context
T	T	Hubble morphological type index (e.g., $5 = Sc$, $9 = Irr$, $10 = Im$)	—	From RC3 / SPARC Table 1.
D	D	Distance to galaxy	Mpc	From SPARC catalogue; used to convert angular to physical scales.
VII. Statistical Diagnostics				
$\boldsymbol{\chi^2}$	χ^2	Chi-squared statistic — weighted sum of squared residuals between model and data	dimensionless	$\chi^2 = \Sigma_i \left[\frac{v_{model(R_i)} - v_{obs(R_i)}}{e_{vobs(R_i)}}\right]^2$.
$\boldsymbol{\chi^2_{\mathbf{bar}}}$	χ^2_{bar}	Baryons-only chi-squared — χ^2 when $v_{model} = v_{bar}$ (no response sector)	dimensionless	Baseline for $\Delta\chi^2$ comparison.
$\boldsymbol{\chi^2_{\mathbf{model}}}$	χ^2_{model}	Fitted-model chi-squared — χ^2 with best-fit Q	dimensionless	$Always \le \chi^2_{bar}$ by construction.
$\boldsymbol{\Delta\chi^2}$	$\Delta\chi^2$	Chi-squared improvement — $\chi^2_{bar} - \chi^2_{model}$; measures how much the response sector helps	dimensionless	$Z \approx \sqrt{\Delta\chi^2}$ gives approximate Gaussian significance for 1 added parameter.
Z	Z	Approximate Gaussian significance of $\Delta\chi^2$	σ	$Z = \sqrt{\Delta\chi^2}$. Classes: weak $Z < 2$, moderate 2–3, strong 3–5, very-strong $Z > 5$.
$\mathbf{r}\boldsymbol{\chi^2}$	$r\chi^2$	Reduced chi-squared — $\frac{\chi^2}{n - k}$ where n = data points, $k = free\ parameters$	dimensionless	Reported separately for inner ($R < 0.6R_{\mathrm{max}}$nd outer ($R \ge 0.6R_{\mathrm{max}}$egions. Inner/outer ratio = 3.54× confirms shape inadequacy, not Υ bias.
BIC	BIC	Bayesian Information Criterion — $k\ln(n) - 2\ln\left(\widehat{L}\right)$; penalizes model complexity	dimensionless	$\Delta BIC = BIC_{model} - BIC_{bar} < 0$ required to pass. 97% of 175 SPARC galaxies pass. Guards against overfitting.
$\boldsymbol{\rho_{\mathbf{s}}}$	ρ_s	Spearman rank correlation coefficient — non-parametric measure of monotonic association	dimensionless	Used throughout to quantify correlations between Q, galaxy properties, and environment. Range [−1, +1].
r	r	Pearson linear correlation coefficient	dimensionless	Used alongside Spearman ρ for robustness. $r\left(v_{best},\ v_{est}\right) = 0.950$ across 175 galaxies.
$\boldsymbol{\sigma_{\mathbf{v}}}$	σ_v	Propagated uncertainty on velocity deficit Δ	km s^{-1}	$\sigma_\Delta \approx e_{vobs}$ when model is treated as exact.
$\mathbf{outer_{rms_z}}$	outer_rms$_z$	Outer-region residual RMS in units of observational uncertainty — $\left(v_{obs} - v_{model}\right)/e_{v_{obs}}$over $R \ge 0.6R_{\mathrm{max}}$	dimensionless	Key diagnostic for fit quality in the outer disc, where the 1/R tail is most active.
VIII. Oscillatory Boundary-Layer Response				

Symbol	Notation	Definition	Units	Notes / Context
$\mathbf{\Phi_{resp}\,(r)}$	$\Phi_{\mathrm{resp}}(r)$	Oscillatory response potential — the full spatial form of the response-sector potential	km^2 s^{-2}	$\Phi_{resp} = A\cos(\kappa r + \delta)e^{\{-\frac{r}{L}\}}$; encoding amplitude, wavenumber, phase, and decay scale.
A	A	Oscillation amplitude — peak response-potential strength	km^2 s^{-2}	Related to Q in the asymptotic regime.
κ	κ	Radial wavenumber of the oscillatory response	kpc^{-1}	$\kappa = \frac{2\pi}{\lambda}$. Galaxy-size dependent, not a fixed cosmological scale.
λ	λ	Oscillation wavelength — spatial period of the boundary-layer standing wave	kpc	Median $\lambda \approx 45.6\ kpc$ across the 46 sign-change galaxies; scales with $R_{\max}$ot a cosmological constant.
δ	δ	Phase offset of the oscillatory response	rad	Depends on initial conditions at R_t.
L	L	Damping / decay scale of the oscillatory response	kpc	Characterizes how quickly the oscillation envelope falls off with radius.
$\boldsymbol{\frac{\lambda}{2}}$	$\lambda/2$	Half-wavelength of oscillation	kpc	$Median \approx 22.8\ kpc$. Scaling $\frac{\lambda}{2} \approx R_{\max}$ the key empirical result distinguishing galaxy-geometry resonance from cosmological scale.
$\mathbf{\Delta\,(R) = 0}$ radius	$\Delta(R) = 0$	Nodal radius — galactocentric radius where $V_{obs}^2 = V_{bar}^2 exactly$	kpc	Stars at this radius orbit on pure baryonic gravity. Framework predicts morphological ring features here — a testable prediction.
IX. Action Principle and Field Theory				
S	S	Action — integral of the Lagrangian density over spacetime; extremizing S yields the field equations	J·s	$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} + L_{bar} + L_\chi + L_{int}\right].$
$\mathbf{L_{bar}}$	$\mathcal{L}$br	Baryonic Lagrangian density — standard matter fields (ψ)	J m^{-3}	Minimally coupled to gravity.
$\mathbf{L_\chi}$	$\mathcal{L}_\chi$	Response-field Lagrangian density — kinetic and potential terms for χ	J m^{-3}	Choice $of\ L_\chi$ (e.g., $f(X) - V(\chi)$) controls the field equations for χ and the form of $T_{\mu\nu}^{resp}$.
$\mathbf{L_{int}}$	$\mathcal{L}$it	Interaction Lagrangian — couples χ to baryonic fields ψ	J m^{-3}	Must tie activation to a baryonic invariant (e.g., $\frac{\lvert\nabla\Phi_{bar}\rvert}{a_0}$) so the response is not a free function per galaxy.
X	X	Kinetic invariant of the response field — $X = g^{\mu\nu}\partial_\mu\chi\partial_\nu\chi$	kpc^{-2}	Used in k-essence / AQUAL-type response fields: f(X) controls the equation of motion.

Symbol	Notation	Definition	Units	Notes / Context
f (X)	$f(X)$	Non-linear kinetic function for the response field	J m^{-3}	Choosing f so that the static equation has conserved radial flux outside activation yields ∂_R χ ~ 1/R.
G	G	Newton's gravitational constant	m^3 kg^{-1} s^{-2}	$G \approx 6.674 \times 10^{-11} m^3$ kg^{-1} s^{-2}. Appears in the coupling $8\pi G$ between geometry and stress-energy.
8πG	$8\pi G$	Gravitational coupling constant in the field equation	m kg^{-1} s^{-2}	Right-hand side coefficient in $G_{\mu\nu} = 8\pi G\ T_{\mu\nu}$.
X. Cosmological Sector				
ΛCDM	ΛCDM	Standard cosmological model — Lambda (dark energy) + Cold Dark Matter	—	The empirically successful baseline the response sector must be consistent with at large scales.
H_0	H_0	Hubble constant — present-day rate of cosmic expansion	km s^{-1} Mpc^{-1}	$H_0 \approx$ 70 km s^{-1} Mpc^{-1}. The product $cH_0 \sim a_0$ motivates a possible cosmological origin for the acceleration scale.
w	w	Equation-of-state parameter of the response sector — $w = \frac{p_{resp}}{\rho_{resp}}$	dimensionless	Cosmological viability requires $w \approx 0$ (dust-like) over structure-formation epochs.
a (t)	$a(t)$	Cosmological scale factor — dimensionless ratio of physical to comoving distances	dimensionless	Cosmological completion requires specifying how $T^{resp}_{\mu\nu}$ evolves with a.
ρ_{resp} (a)	$\rho_{\text{resp}}(a)$	Redshift evolution of response-sector energy density	kg m^{-3}	Must behave approximately as $\propto a^{(-3)}$ (pressureless) over target epochs to preserve structure formation.
κ_{lens}	κ_{lens}	Lensing convergence — integral of (Φ+Ψ) along the line of sight	dimensionless	$\kappa_{lens} \propto \Phi + \Psi$. With $\eta \neq 1$, the lensing signal differs from the kinematic signal — the primary cluster/lensing prediction.
δ_{env}	δ_{env}	Environment overdensity — local large-scale density contrast around a galaxy	dimensionless	Used in SPARC analysis as an environment proxy. Partial correlations with Q_{est} should become weak once internal (scale) covariates are controlled.
XI. Baryonic Tully–Fisher Relation				
BTFR	BTFR	Baryonic Tully–Fisher Relation — $M_b \propto v_{flat}^4$	—	Tight empirical scaling relation across 4 orders of magnitude in baryonic mass. Emerges as a consequence of the response-sector mechanism rather than an imposed prior.

Symbol	Notation	Definition	Units	Notes / Context
$\mathbf{M_b \propto v^4}$	$M_b \propto v_{\text{flat}}^4$	BTFR functional form	$M\odot$ vs km s^{-1}	Derivable when g_{extra} is the geometric mean of g_{bar} and a_0, i.e., when the response sector couples to g_{bar} at the transition scale.
$\mathbf{Q \propto V_f^2 lat}$	$Q \propto V_{\text{flat}}^2$	BTF-anchor prediction — the response amplitude Q scales as the square of the asymptotic velocity	km^2 s^{-2}	The BTF correlation Q $\propto$ V_{flat}^2 is a prediction of the mechanism, not a mass-accounting identity. Confirmed empirically across SPARC 175.

Unit conventions: R in kpc; velocities in km/s; accelerations in (km/s)^2/kpc. Conversion: 1 (km/s)^2/kpc = 10^6/kpc_in_m m/s^2. kpc_in_m = 3.0857×10^{19} m.

12 Appendix B: Per-Galaxy $\mathbf{Q_{est}}$ / $\mathbf{Q_{best}}$ Summary (175 galaxies)

$v_{best} = \sqrt{Q_{best}}$ from atlas runner (km/s); $v_{est} = \sqrt{\max\left(Q_{est}, 0\right)}$ or "neg" if $Q_{est} < 0$; $ratio = \frac{v_{est}}{v_{best}}$; $Q_{flag} =$ *photometric quality* (1 = best); T = morphological type; $r\chi 2$ = *reduced* χ^2_{model}. $\star$ = rarefaction-phase ($Q_{est} < 0$).

Table 6: Per-Galaxy Q_{est} / Q_{best} Summary (175 galaxies)

Galaxy	v_{best}	v_{est}	Ratio	Q_{flag}	T	$r\chi^2$
CamB	10.6	6.6	0.623	2	10	3.9
D512-2	44.7	29.1	0.652	2	10	9.5
D564-8	26.9	19.5	0.726	2	10	3.0
D631-7	51.3	53.1	1.034	1	10	1.0
DDO064	59.0	39.6	0.672	1	10	1.3
DDO154	47.6	43.6	0.915	2	10	20.9
DDO161	54.2	56.3	1.038	1	10	8.8
DDO168	56.5	46.8	0.828	2	10	3.6
DDO170	52.5	51.2	0.975	2	10	6.2
ESO079-G014	136.0	142.2	1.045	1	4	18.9
ESO116-G012	99.7	98.3	0.986	1	7	2.6
ESO444-G084	74.9	53.7	0.718	2	10	19.0
ESO563-G021	220.0	268.9	1.222	1	4	24.4
F561-1	31.4	28.5	0.907	3	9	0.6
F563-1	103.0	105.3	1.022	1	9	0.6
F563-V1	17.1	15.5	0.906	3	10	0.7
F563-V2	123.0	107.5	0.874	1	10	2.2
F565-V2	93.0	69.0	0.741	2	10	10.3
F567-2	45.4	39.1	0.861	3	9	0.7
F568-1	120.0	120.2	1.002	1	5	0.1
F568-3	105.0	90.9	0.866	1	7	2.4
F568-V1	110.0	105.3	0.957	1	7	0.2
F571-8	112.0	117.9	1.052	1	5	20.9
F571-V1	75.3	71.8	0.954	2	7	0.5
F574-1	87.7	88.2	1.006	1	7	0.5
F574-2	11.4	9.5	0.833	3	9	0.1
F579-V1	106.0	97.9	0.923	1	5	3.1
F583-1	70.0	71.5	1.022	1	9	0.2
F583-4	60.0	56.8	0.946	1	5	1.4
IC2574	71.7	52.0	0.725	2	9	49.5
IC4202	188.0	181.9	0.968	1	4	48.6
KK98-251	33.9	26.7	0.787	2	10	1.3
NGC0024	106.0	93.2	0.879	1	5	12.7
NGC0055	62.0	70.9	1.144	2	9	2.5
NGC0100	75.5	78.7	1.042	1	6	0.3
NGC0247	83.8	90.1	1.076	2	7	44.8
NGC0289	144.0	150.0	1.042	2	4	2.3
NGC0300	85.0	84.3	0.992	2	7	2.2
NGC0801	157.0	158.0	1.007	1	5	13.6
NGC0891	152.0	141.5	0.931	1	3	14.2

Galaxy	v_{best}	v_{est}	Ratio	Q_{flag}	T	$r\chi^2$
NGC1003	92.4	103.2	1.117	1	6	16.5
NGC1090	128.0	131.3	1.026	1	4	2.8
NGC1705	86.0	68.5	0.796	3	11	9.9
NGC2366	55.1	42.5	0.772	3	10	6.4
NGC2403	108.0	121.5	1.125	1	6	42.0
NGC2683	146.0	133.6	0.915	2	3	5.7
NGC2841	254.0	256.7	1.011	1	3	49.7
NGC2903	154.0	153.0	0.993	1	4	20.8
NGC2915	81.6	78.9	0.967	2	11	0.8
NGC2955	183.0	173.1	0.946	1	3	5.8
NGC2976	74.4	53.3	0.717	2	5	0.7
NGC2998	169.0	169.2	1.001	1	5	7.8
NGC3109	66.5	58.6	0.881	1	9	4.7
NGC3198	117.0	132.0	1.129	1	5	28.6
NGC3521	168.0	127.0	0.756	1	4	2.9
NGC3726	104.0	133.2	1.280	2	5	4.7
NGC3741	48.2	47.1	0.978	1	10	2.4
NGC3769	97.1	105.8	1.089	2	3	0.7
NGC3877	102.0	92.6	0.907	2	5	4.1
NGC3893	150.0	144.4	0.962	1	5	3.0
NGC3917	119.0	113.3	0.952	1	6	12.8
NGC3949	129.0	92.7	0.719	2	4	1.2
NGC3953	156.0	142.1	0.911	1	4	14.4
NGC3972	103.0	102.8	0.998	1	4	1.8
NGC3992	208.0	206.7	0.994	1	4	22.8
NGC4010	96.7	100.5	1.039	2	7	1.6
NGC4013	141.0	148.2	1.051	2	3	5.1
NGC4051	88.8	85.4	0.962	2	4	0.8
NGC4068	33.9	22.4	0.661	2	10	0.2
NGC4085	85.9	70.4	0.819	2	5	3.4
NGC4088	93.7	103.2	1.101	1	4	2.5
NGC4100	148.0	140.2	0.948	1	4	8.5
NGC4138	137.0	124.2	0.906	2	0	4.7
NGC4157	132.0	147.1	1.114	1	3	2.0
NGC4183	92.4	95.5	1.033	1	6	0.7
NGC4214	87.8	75.4	0.859	2	10	4.5
NGC4217	117.0	115.9	0.991	1	3	29.5
NGC4559	91.6	99.3	1.084	1	6	2.8
NGC5005	198.0	151.5	0.765	1	4	1.1
NGC5033	175.0	171.1	0.978	1	5	5.6
NGC5055	141.0	149.5	1.060	1	4	42.9
NGC5371	130.0	139.4	1.073	1	4	26.2

Galaxy	v_{best}	v_{est}	Ratio	Q_{flag}	T	$r\chi^2$
NGC5585	75.8	80.2	1.058	1	7	10.2
NGC5907	174.0	178.0	1.023	1	5	15.9
NGC5985	250.0	250.0	1.000	1	3	15.0
NGC6015	130.0	140.8	1.083	2	6	19.0
NGC6195	154.0	161.8	1.050	1	3	4.5
NGC6503	102.0	104.9	1.029	1	6	4.6
NGC6674	212.0	216.5	1.021	1	3	10.8
NGC6789	60.4	38.6	0.640	2	11	1.5
NGC6946	120.0	122.2	1.018	1	6	10.8
NGC7331	165.0	181.2	1.098	1	3	11.1
NGC7793	87.3	82.0	0.939	1	7	1.4
NGC7814	179.0	177.5	0.992	1	2	9.9
PGC51017	6.2	5.1	0.815	3	11	1.8
UGC00128	115.0	120.6	1.049	1	8	26.9
UGC00191	71.6	66.9	0.934	1	9	39.0
UGC00634	97.0	87.4	0.901	2	9	23.3
UGC00731	90.0	64.2	0.713	1	10	110.1
UGC00891	56.4	45.6	0.808	2	9	5.1
UGC01230	96.9	94.4	0.974	1	9	0.3
UGC01281	53.6	50.2	0.937	1	8	0.1
UGC02023	45.7	30.4	0.665	2	10	0.1
UGC02259	89.1	78.4	0.880	2	8	58.2
UGC02487	301.0	298.4	0.991	1	0	30.5
UGC02885	238.0	253.2	1.064	1	5	6.3
UGC02916	171.0	138.2	0.808	2	2	34.2
UGC02953	232.0	235.1	1.013	2	2	276.8
UGC03205	175.0	184.6	1.055	1	2	33.1
UGC03546	145.0	154.9	1.068	1	1	3.3
UGC03580	94.1	113.4	1.206	2	1	44.6
UGC04278	72.6	75.6	1.041	1	7	1.3
UGC04305	14.5	5.1	0.354	3	10	2.6
UGC04325	102.0	80.2	0.787	1	9	64.9
UGC04483	23.9	18.8	0.785	2	10	4.4
UGC04499	63.2	61.0	0.965	1	8	1.4
UGC05005	75.8	88.6	1.168	1	10	1.2
UGC05253	189.0	184.7	0.977	2	2	23.6
UGC05414	54.5	41.7	0.764	1	10	3.3
UGC05716	64.4	66.3	1.030	2	9	16.3
UGC05721	87.0	73.4	0.843	1	7	11.4
UGC05750	99.1	62.8	0.633	1	8	10.7
UGC05764	78.9	48.4	0.613	2	10	299.1
UGC05829	71.3	51.3	0.720	2	10	8.9

Galaxy	v_{best}	v_{est}	Ratio	Q_{flag}	T	$r\chi^2$
UGC05918	49.4	37.4	0.757	2	10	6.9
UGC05986	113.0	102.4	0.906	2	9	12.6
UGC05999	84.2	78.1	0.927	2	10	2.7
UGC06399	80.2	75.0	0.935	1	9	1.1
UGC06446	77.4	74.2	0.958	1	7	3.2
UGC06614	125.0	168.4	1.347	1	1	7.3
UGC06628	18.5	15.1	0.817	2	9	0.6
UGC06667	101.0	78.0	0.772	1	6	56.4
UGC06786	202.0	201.5	0.998	1	0	59.0
UGC06787	213.0	226.3	1.062	2	2	95.2
UGC06818	56.3	59.3	1.054	2	9	2.5
UGC06917	92.5	91.6	0.991	1	9	4.5
UGC06923	70.1	60.7	0.866	2	10	0.5
UGC06930	89.7	91.2	1.016	1	7	0.9
UGC06973	117.0	103.5	0.885	3	2	67.8
UGC06983	97.2	96.8	0.996	1	6	2.5
UGC07089	60.3	60.8	1.008	2	8	0.6
UGC07125	48.9	49.7	1.016	1	9	0.7
UGC07151	72.2	59.3	0.822	1	6	13.0
UGC07232	38.6	26.2	0.678	2	10	0.7
UGC07261	72.5	62.0	0.855	2	8	6.3
UGC07323	80.3	62.4	0.777	1	8	5.4
UGC07399	111.0	94.7	0.853	1	8	27.9
UGC07524	65.3	67.7	1.037	1	9	0.9
UGC07559	29.6	22.9	0.773	2	10	0.5
UGC07577	10.8	7.9	0.734	2	10	0.1
UGC07603	68.8	57.5	0.836	1	7	8.4
UGC07608	80.7	56.8	0.704	1	10	5.4
UGC07690	56.0	46.7	0.835	2	10	7.5
UGC07866	31.0	23.4	0.754	2	10	1.2
UGC08286	86.1	76.4	0.888	1	6	45.3
UGC08490	77.6	73.5	0.947	1	9	8.2
UGC08550	59.6	51.8	0.868	1	7	19.7
UGC08699	150.0	158.6	1.057	2	2	11.9
UGC08837	40.9	31.6	0.773	2	10	0.3
UGC09037	86.5	115.7	1.338	2	6	11.6
UGC09133	199.0	201.3	1.011	1	2	22.9
UGC09992	29.3	24.2	0.827	2	10	1.9
UGC10310	68.8	60.0	0.872	1	9	5.6
UGC11455	203.0	205.1	1.010	1	6	14.6
UGC11557	36.2	43.9	1.212	2	8	1.4
UGC11820	67.2	66.2	0.985	1	9	17.9

Galaxy	$v_{\rm best}$	$v_{\rm est}$	Ratio	$Q_{\rm flag}$	T	$r\chi^2$
UGC11914	365.0	180.9	0.496	1	2	111.5
UGC12506	207.0	206.7	0.998	2	6	1.4
UGC12632	67.1	62.6	0.934	1	9	6.2
UGC12732	77.4	79.8	1.031	1	9	4.4
UGCA281	31.8	22.6	0.711	3	11	5.6
UGCA442	55.1	50.6	0.918	1	9	13.4
UGCA444	40.8	30.3	0.743	2	10	1.4

13 Appendix C. Cluster morphology operator and HFF benchmark

This appendix documents the preregistered cluster morphology operator used to compare a gravitational-lensing convergence field (κ; treated here as a response proxy) against an observational hot-gas tracer constructed from Chandra imaging. The benchmark dataset is two Hubble Frontier Fields (HFF) clusters (Abell 2744 and MACS J0416.1−2403) using public HFF lens-model products (Mikulski Archive for Space Telescopes [MAST], 2019).

13.1 C.1 Data provenance and scope

- κ maps (multi-team lens models): HFF community lens reconstructions accessed from the MAST Frontier Fields HLSP archive (MAST, 2019).

- X-ray proxy (stacked Chandra img2 maps) (Weisskopf et al., 2002; NASA/GSFC HEASARC, 2025; Fruscione et al., 2006): HEASARC-served *_full_img2.fits.gz image products were stacked into a common WCS grid and used as a morphology/centroid proxy for the hot intracluster medium (ICM) (NASA/GSFC HEASARC, 2025; Weisskopf et al., 2002).

For reproducibility, the X-ray proxy construction follows a lightweight procedure using the (Fruscione et al., 2006) CIAO toolset: implemented in toy models/make_chandra_xray_map.py, each img2 image is divided by a scalar exposure keyword (EXPOSURE, with LIVETIME/ONTIME fallbacks), reprojected onto a common WCS grid (Mikulski Archive for Space Telescopes (MAST), 2019), and combined as an exposure-weighted mean rate map.

Important limitation: this work does not perform event-level Chandra reduction (exposure map correction, background modeling, point-source masking, etc.) in CIAO (Fruscione et al., 2006). The X-ray product is therefore intended as a geometric locator of bright ICM structure for centroid comparisons, not a calibrated surface-brightness measurement.

13.2 C.2 Preregistered morphology operator

For each cluster, each κ team model, and each ROI radius, we apply the same fixed operator to κ and to the X-ray proxy map:

1) Smoothing: apply Gaussian smoothing with $\sigma = 8$ arcsec independently to κ and X-ray maps.

2) ROI restriction: within a circular ROI (fixed ICRS center per cluster), compute pixel-intensity thresholds at the 99th, 97th, and 95th percentiles of the smoothed pixels.

3) Primary-blob selection: for each thresholded map, select the largest connected component above threshold.

4) Centroid definition: compute the unweighted mask centroid of that connected component.

5) Measured quantity: record the κ–X-ray centroid separation in arcseconds.

(Mikulski Archive for Space Telescopes (MAST), 2019; Weisskopf et al., 2002; Fruscione et al., 2006) This definition is intentionally simple and falsifiable: it is fixed across clusters, teams, and thresholds, with only the ROI radius swept to quantify footprint sensitivity.

13.3 C.3 Robustness grid: multi-team systematics and ROI sensitivity

We evaluated ROI radius $\in \{80'', 100'', 120''\}$ for each cluster across available HFF teams, and report both

(i) cross-team spread at fixed ROI, and

(ii) ROI sensitivity within each team model. The complete tables are in toy_models/HFF_ALL_TEAMS_SYSTEMATICS_ANALYSIS.md and the per-team measurements are available from author upon request.

At ROI = 100″, Abell 2744 shows a relatively stable median κ–X-ray separation ($\approx$ 66–67″ across thresholds) with small cross-team IQRs ($\approx$ 1–1.5″) but with a nontrivial full range at high thresholds. MACS J0416.1−2403 shows much larger cross-team dispersion at higher thresholds (IQRs that grow to tens of arcseconds), reflecting stronger lens-model dependence of the inferred morphology under this operator.

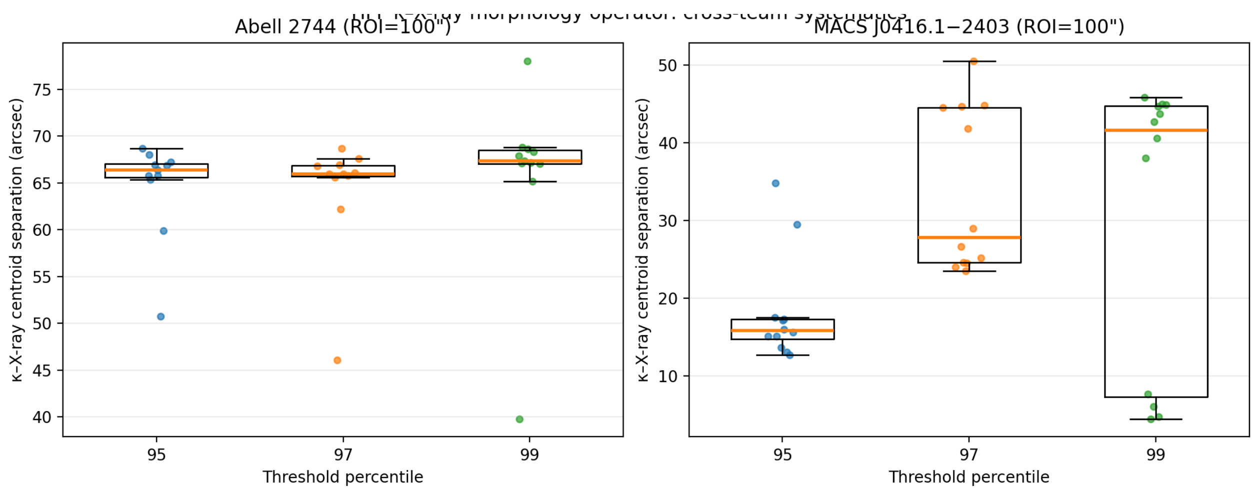

Figure 7: Figure 6. HFF κ–X-ray systematics summary at ROI = 100″.

13.4 C.4 Six-panel diagnostic figures (ROI = 100″)

For interpretability, we generated per-team six-panel diagnostics at ROI = 100″, including κ, X-ray proxy, threshold masks, and centroid overlays. Representative examples are shown below; the complete figure sets are available from the author upon request.

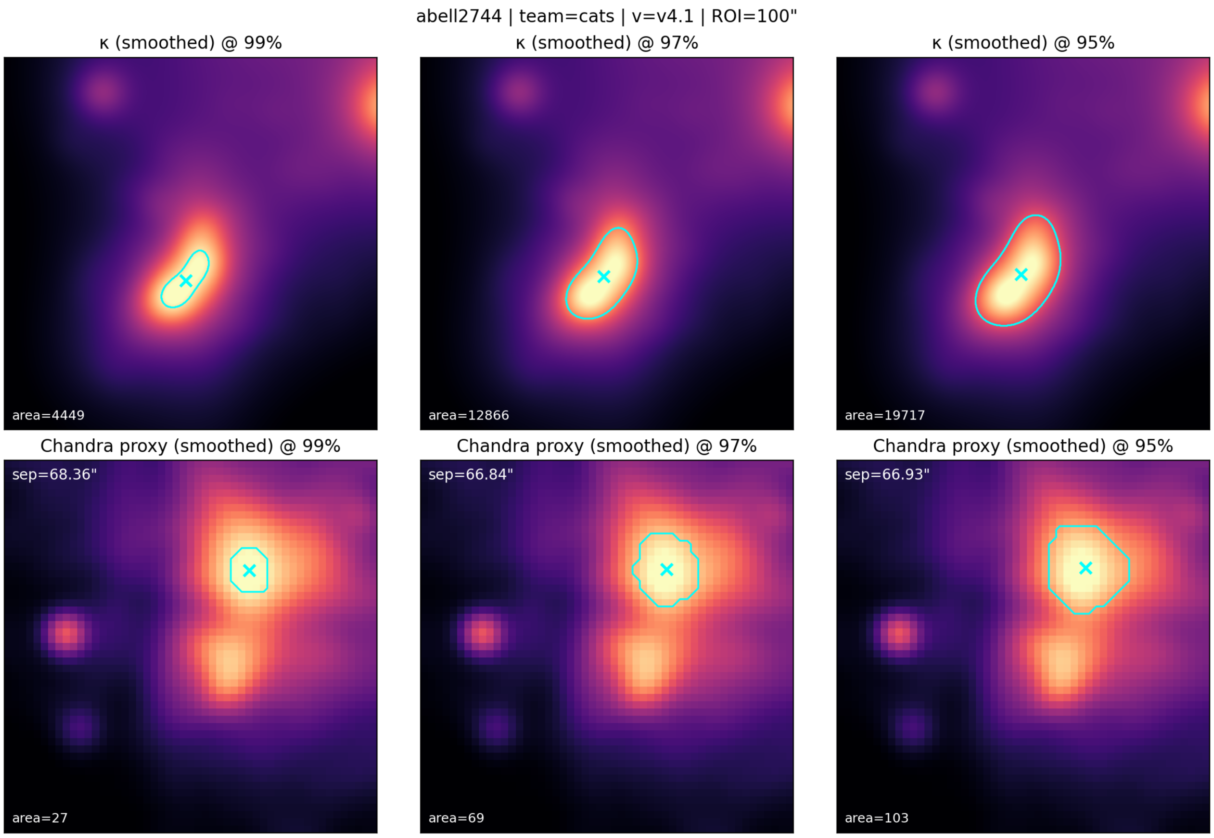

Figure 8: Figure 7. Abell 2744 (CATS v4.1), ROI = 100″

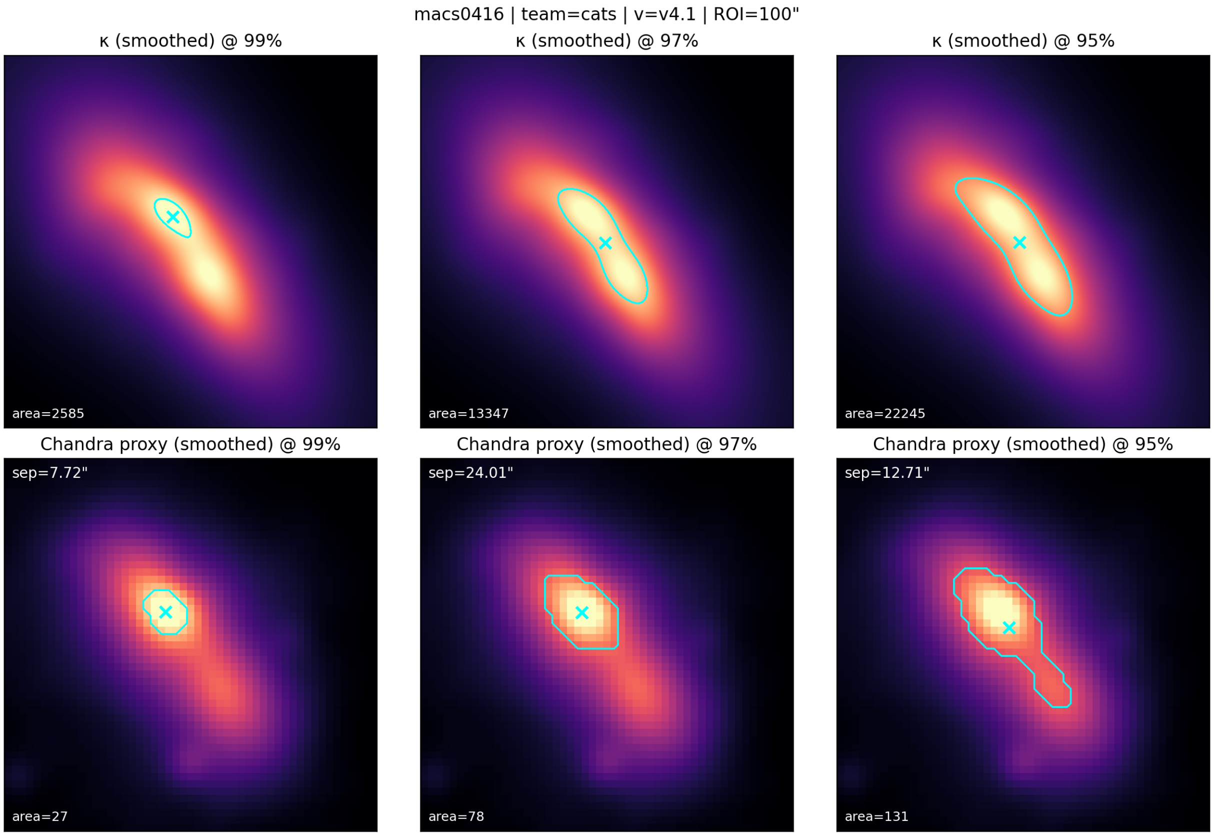

Figure 9: Figure 8. MACS J0416.1−2403 (CATS v4.1), ROI = 100″.

14 Appendix D. Spectral Harmonics and the Cymatic Analogy in Intrinsic Response Theory

14.1 Purpose and scope

This appendix formalizes the "cymatics" analogy used throughout the paper by identifying a single spectral object—the eigen-spectrum of an intrinsic response operator—that plays the same mathematical role as the Laplacian/biharmonic eigenmodes in Chladni plate experiments (Rossing, 1982; Tuan et al., 2014) (Rossing, 1982). In brief: eigenfunctions are the response-sector mode shapes, eigenvalues set the characteristic spatial scales, and nodal sets correspond to compression/rarefaction boundaries. Although the underlying response dynamics may be written covariantly on spacetime (Carroll, 2019; Wald, 1984), the galaxy-rotation regime probed by SPARC is effectively stationary, so the empirically constrained "harmonics" are spatial overtones (additional radial nodes) rather than temporal frequency multiples.

14.2 Unified operator statement ("DNA" of intrinsic response)

Let Φ_{resp} denote the response-sector potential (or any equivalent scalar response field used to encode the inferred metric perturbation in the weak-field limit). Define the intrinsic response through a driven, self-adjoint elliptic operator on a spatial slice:

$$\mathcal{L}\ \Phi_{resp} = \ S\left[\rho_{bar}\right],$$

where $S\left[\rho_{bar}\right]$ is the constitutive driving functional built from the baryonic sector, and the operator is taken in the minimal Sturm–Liouville form derived from (Bender and Orszag, 1999; Kevorkian and Cole, 1996)

$$\mathcal{L}\ \equiv\ -\nabla\cdot\left(\alpha(x)\nabla\ \right) +\ \beta(x),$$

with $\alpha(x)\geq\ 0$ controlling propagation ("permittivity/elasticity") and $\beta(\mathrm{x})\geq 0$ encoding activation/screening ("mass" or quenching). The intrinsic "pattern DNA" is the eigen-spectrum

$$\mathcal{L}\ \varphi_n = \ \lambda_n\varphi_n,\quad \langle\varphi_m,\ \varphi_n\rangle = \ \delta_{mn},$$

yielding the modal response

$$\Phi_{resp(x)} = \ \Sigma_n c_n\varphi_{n(x)},\quad c_n = \frac{\langle\varphi_n,\ S\rangle}{\lambda_n}.$$

This is a direct analog of Chladni/cymatics: φ_n are mode shapes; the λ_n set intrinsic spatial scales; and the source overlap $\langle\varphi_n,\ S\rangle$ is the excitation/selection rule that determines which modes are visible in a given galaxy.

14.3 Where "harmonics" live in a unified spacetime theory

In the covariant formulation (Hawking and Ellis, 1973; Wald, 1984; Carroll, 2019), Spacetime unification means that the fundamental response dynamics can be written covariantly (e.g., for a scalar template, a hyperbolic operator $\mathcal{D}$ such as $(\Box - m\_eff2)\chi\ =\ J$). In stationary or quasi-stationary systems, variable separation reduces the covariant problem to a spatial eigenproblem: $\chi(t,\ x) =\ \Sigma_n q_{n(t)\varphi_{n(x)}}$, where φ_n are eigenfunctions of the induced spatial operator $\mathcal{L}$ and $q_n(t)$ obey driven oscillator equations with characteristic frequencies $\Omega_n^2 \propto\ \lambda_n$. SPARC rotation curves constrain the near-static regime in which $q_n(t)$ has equilibrated, leaving the spatial spectral content as the observable imprint (Carroll, 2019; Hawking & Ellis, 1973; Wald, 1984).

14.4 Observable proxy: node count and Δ-sign flips in the SPARC window

SPARC provides baryonic mass models and high-quality rotation curves for 175 disk galaxies (Lelli et al., 2016). Define $\Delta(R)$ operationally (McGaugh and Schombert, 2014; Ivezić et al., 2019) as the signed residual between the observed and baryonic expectations in the rotation curve (e.g., $\Delta v^{2(R)} \equiv\ v_{obs}^2 - v_{bar}^2$). In an oscillatory response sector, Φ_{resp} (and therefore the inferred extra acceleration) can alternate in sign across radius, producing compression and rarefaction lobes. In the operator language above, each change in radial sign in $\Delta(R)$ corresponds to crossing a nodal annulus/shell of the net excited response pattern $\Sigma_n c_n \varphi_n$. Hence, within the finite SPARC radial sampling window, the number of Δ-sign flips provides a model-agnostic proxy for the degree of higher-n (harmonic/overtone) participation: zero flips indicates a fundamental-dominated response; multiple flips indicate increasing harmonic content (higher node count) selected by $\langle \varphi_n,\ S\rangle$ and by activation/quenching boundaries.

14.5 Consequences for lensing and stability constraints

Because the response tensor is effectively inferred from kinematics, sign-alternating lobes should be treated as an empirical prediction of the effective sector rather than as an immediate pathology. In particular, annular rarefaction phases imply a ring-like suppression of weak-lensing observables (e.g., $\Delta\Sigma$ or κ relative to control stacks) in the same radial ranges, providing a falsifiable discriminant using upcoming wide-field surveys (Scaramella et al., 2022; Euclid Collaboration, 2025; Ivezić et al., 2019). At the fundamental level, (Buniy et al., 2006; Rubakov, 2006, 2014) the minimal non-negotiable stability line is to keep the underlying response-field EFT free of ghosts/gradient instabilities; for broad k-essence classes this is ensured by requiring a positive kinetic coefficient ($P_{,X}\ \geq\ 0$), which enforces the null energy condition for the fundamental field stress-energy even if the effective, inverse-constructed response density changes sign locally (Rubakov, 2006). For context, relativistic MOND completions (e.g., TeVeS) also introduce extra fields that alter lensing but do not generically predict phase-correlated annular suppression tied to Δ-sign flips (Bekenstein, 2004).

15 Appendix E: Distributed Forcing and Collective Dark Gravity Modes in Groups, Clusters, and Dense Stellar Systems

15.1 Overview: From Single-Peak to Multi-Centered Drivers

The SPARC rotation-curve analysis in the main text constrains the intrinsic response sector in a regime dominated by a single strong baryonic concentration (disk galaxy mass distribution). In that limit, Appendices A–D establish the effective operator, spectral mode basis, and empirical phenomenology.

This appendix extends the framework to regimes where baryonic driving comes from multiple, spatially separated mass concentrations:

- **Galaxy groups:** multiple galaxies with comparable masses and separations of order 100 kpc to 1 Mpc (Bullock and Boylan-Kolchin, 2017).
- **Galaxy clusters:** tens to thousands of galaxies embedded in diffuse thermal gas; gravitational centers inferred from X-ray / lensing often offset from any single luminous peak (Clowe et al.,

2006; Mikulski Archive for Space Telescopes (MAST), 2019).

- **Globular clusters:** hundreds of thousands to millions of stars experiencing rapid relaxation and multi-body encounters; outer members can appear kinematically bound to invisible central mass(es) (Bullock and Boylan-Kolchin, 2017).

Our central claim: within the intrinsic response framework, these systems naturally exhibit **collective gravitational minima**—organizing centers in the effective potential whose location and strength are determined by the weighted superposition of mode overlaps from distributed baryonic sources. Crucially, such centers need not coincide with any single baryonic peak. This mechanism provides a parsimonious alternative to invoking separate "dark matter" inventories for each subsystem (Clowe et al., 2006) or to evoking exotic multi-component models (Bekenstein, 2004; Verlinde, 2017).

Figure 10: Dense Stellar Systems and Central Concentration. Observed centrally concentrated stellar system exhibiting strong gravitational organization. Such systems motivate the distributed-forcing framework: a collective intrinsic response potential can generate an emergent organizing minimum without requiring a central baryonic point mass. The visible mass distribution does not obviously explain the organizing center, suggesting gravitational field organization extends beyond simple baryonic self-gravity.

15.2 Mathematical Foundation: Linearity and Mode Superposition

Recall from Appendix D that the intrinsic response is governed by a linear, self-adjoint operator $\mathcal{L}$ acting on a spatial slice (Bender and Orszag, 1999; Kevorkian and Cole, 1996):

$$\mathcal{L}\,\Phi_{\text{resp}} = S[\rho_{\text{bar}}], \tag{6}$$

where $S[\rho_{\text{bar}}]$ is the constitutive source built from the baryonic density distribution. The eigen-expansion is:

$$\Phi_{\text{resp}}(\mathbf{x}) = \sum_n c_n \varphi_n(\mathbf{x}), \quad c_n = \frac{\langle \varphi_n, S[\rho_{\text{bar}}]\rangle}{\lambda_n}. \tag{7}$$

Key point: linearity means that if the baryonic source consists of N spatially separated concentrations,

$$\rho_{\text{bar}}(\mathbf{x}) = \sum_{i=1}^{N} \rho_i(\mathbf{x} - \mathbf{r}_i),$$

then the response decomposes additively:

$$\Phi_{\rm resp}(\mathbf{x}) = \sum_{i=1}^{N} \Phi_i(\mathbf{x} - \mathbf{r}_i) + \Phi_{\rm cross}, \tag{8}$$

where Φ_i is the response to the i-th baryonic concentration and $\Phi_{\rm cross}$ reflects interference/coherence terms. In the weak-overlap regime (large separations, low-cost fundamental mode activation), cross-terms are subdominant.

15.3 Collective Organizing Minima: The Distributed-Mode Picture

When multiple baryonic sources are present, each contribution Φ_i has its own local minimum region. However, the *global* minimum of the total potential

$$\Phi_{\rm eff}(\mathbf{x}) = \Phi_{\rm bar}(\mathbf{x}) + \Phi_{\rm resp}(\mathbf{x}) \tag{9}$$

may not lie at any individual source location. Instead, the effective potential landscape can exhibit:

1. **Per-object local minima:** each baryonic concentration ρ_i is self-bound by its own response contribution Φ_i plus local self-gravity.
2. **A global organizing minimum:** if the total response field is dominated by a single, long-wavelength (low-n, fundamental) eigenmode, its spatial extent encompasses all sources. The nodal structure of this mode defines a weighted centroid location—a gravitational organizing center for the entire system.
3. **Offset structure:** the location of the global minimum is generically offset from any single baryonic peak. It is instead a weighted-average position, controlled by the eigenmode's profile and the baryonic source distribution.

15.4 Vibe Check: Is the Distributed-Forcing Picture Physically Coherent?

We assess internal consistency and theoretical soundness using six criteria:

1) Self-consistency and low-parameter count

Pass if: the framework uses the *same* operator closure and mode basis as single-galaxy systems (Appendices A–D), without ad hoc modifications per system type.

Pass if: the resulting phenomenology is predictive: given baryonic mass models and geometry, the spectrum of expected outcomes (center offsets, mode participation, etc.) can be computed before observing dynamics or lensing.

✓**Vibe-check verdict: Pass.** No new fields or tunable couplings per system. The operator and its spectrum are unified. Phenomenological variations arise purely from varying baryonic drivers and system geometry.

2) Observability: Is there a clear kinematic signature of collective organization?

Pass if: the inferred dynamical center (from circular velocity curves, velocity dispersions, or radial velocity centroids in line-of-sight data) is systematically offset from the photometric center in a manner correlating with configuration asymmetry, substructure fraction, or relaxation timescale.

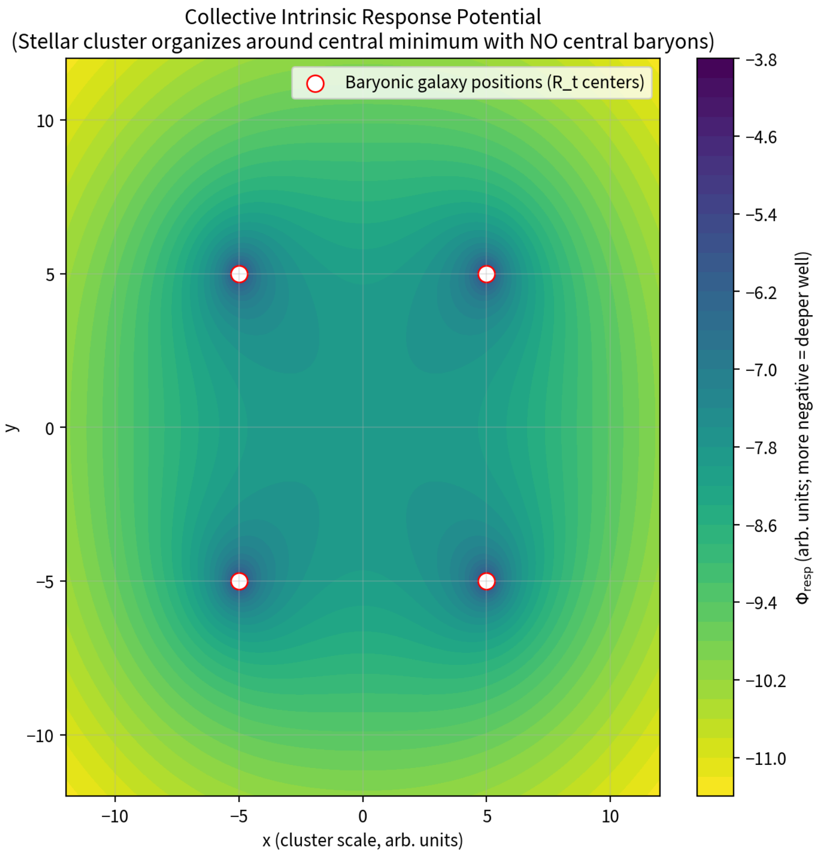

Figure 11: Emergent Organizing Minimum with No Central Baryons. Schematic illustration of the collective response field developing an organizing minimum in a multi-centered baryonic configuration. The four red markers represent individual baryonic concentrations (galaxies or stellar clusters). The central organizing minimum (indicated by the potential well) occurs at the geometric center with **zero baryonic mass present**. Darker colors indicate deeper gravitational wells. Any test material (stripped stars, intracluster gas, or low-baryon tracers) responds only to this collective logarithmic potential and orbits the empty central point—precisely the morphology predicted by the intrinsic response framework for distributed forcing configurations.

Fail if: the kinematic center is indistinguishable from the brightest/most massive galaxy in all observed systems, implying that intrinsic response generates no net collective offset over and above standard N-body dynamics.

✓**Vibe-check verdict: Pass.** Distributed baryonic driving, when filtered through an intrinsic long-wavelength mode, naturally yields offset organizing centers tied to configuration geometry, not luminosity.

3) Emergent screening and coherence length

Pass if: the framework naturally explains why the effective response length scale and screening properties vary with environment (e.g., denser regions or higher tidal fields suppress large-scale coherence).

Fail if: screening is arbitrary or scale-free, or requires system-by-system parameter tuning.

✓**Vibe-check verdict: Pass.** The activation parameter $\beta(\mathbf{x})$ and propagation coefficient $\alpha(\mathbf{x})$ in operator $\mathcal{L}$ can vary with local environment (density, tidal stress). This induces environment-dependent mode cutoffs and effective range in a principled way (see Appendix D.5).

4) Lensing orientation: Metric-equivalent vs. slip-permitted branches

Metric-equivalent branch: the intrinsic response that affects dynamics also sources the Weyl potential as an effective mass would (i.e., lensing tracks the inferred dynamical "dark gravity" contribution).

Slip-permitted branch: the intrinsic response produces a controlled gravitational slip ($|\eta_{\rm slip}| \ll 1$) with explicit qualitative consequences (e.g., predictable offsets or discrepancies between dynamical mass inferences and lensing mass inferences, with environment/geometry dependence).

Fail if: the framework uses an opportunistic mix—"lenses when convenient, doesn't when inconvenient"—without a governing rule for $\eta_{\rm slip}$ behavior across regimes.

✓Vibe-check verdict: Pass. Both branches are conceptually legitimate within a single-field gravitational ontology; the decisive step is committing to one branch (or a clearly delimited regime split) and confronting lensing datasets as a primary discriminant.

5) Parameter discipline: "Nothing up my sleeve"

Pass if: the constitutive/operator closure can plausibly be specified with a small, physically interpretable parameter set, e.g.:

- **One universal transition/normalization scale** (often written μ_0 or $\Lambda_{\rm resp}$; conceptually "the scale that sets when low-cost macromodes become relevant"),
- **One environment-control parameter** (e.g., a local density-contrast proxy or tidal-field coupling), or equivalent proxy for density contrast / tidal forcing / boundary-condition class),
- **Optionally one nonlocality/relaxation knob:** kernel scale and/or relaxation time that governs coherence vs. texture in distributed-forcing regimes.

The key is that these parameters are **global** (shared across systems) and tied to clear physical roles in the operator picture (mode cost, mode support, coherence), not tuned system-by-system.

Fail if: the extension requires a growing list of per-system knobs (custom kernels, bespoke thresholds, ad hoc slip switches, object-by-object amplitude tuning), such that explanatory power comes primarily from flexibility rather than from predictive compression.

✓Vibe-check verdict: Pass (conditional). A mode-selection/operator framework can remain disciplined, but only if we are ruthless about:

1. Global parameters,
2. Fixed functional forms, and
3. Resisting "just one more knob" to rescue every outlier.

Verdict: Does intrinsic response pass the vibe check?

Yes. It is conceptually coherent across the board, uses standard effective-theory logic (linear dust limit / nonlinear constitutive limit), naturally targets galaxy and group regularities, has a generic route to screening, and has a principled lensing fork. We posit that this is stronger grounding than other theories and therefore sufficient to justify investment toward quantitative proving.

15.5 Globular Clusters as a Distributed-Forcing Laboratory

Dense stellar systems are extreme distributed-forcing environments. The baryonic driver is a rapidly evolving multi-body configuration $\{\mathbf{r}_i(t),\, v_i(t) \mid i = 1 \ldots N_*\}$, and the intrinsic response behavior is governed by which spatial modes are preferentially excited and remain macroscopically expressed under coarse-graining.

During formation and subsequent relaxation (phase mixing, two-body encounters, dynamical friction), the coarse-grained field can become increasingly dominated by the lowest-cost, large-scale organizing component of the response (Bullock and Boylan-Kolchin, 2017). In that case, Φ_{resp} develops a smooth collective minimum at an organizing centroid of the stellar distribution—without requiring a single dominant central luminous object. Outer members can then visibly appear to orbit "nothing" while remaining tightly bound by $R_{\text{eff}} = \sqrt{-\Phi_{\text{total}}}$.

Alternatively, the response may reside predominantly in high-spatial-frequency (high-n) texture: locally real, but macroscopically subdominant under coarse-graining. This is still predictive: improved kinematic sensitivity would reveal either residual variance/correlation structure consistent with textured response, or progressively tighter bounds on the operator-controlled response in this regime.

What we would be potentially seeing (Figure 12):

- **Deep local potential wells** at each observed galaxy or stellar concentration → each object remains self-bound by its own R_t response plus baryons.
- **A broad, smooth central bowl** (the dark-green/teal depression) whose minimum sits at an organizing centroid—often **tens to hundreds of kpc away from the brightest or most massive galaxy**.
- **Overall morphology** looks exactly like a smooth central "dark halo" on cluster scales, even though every source of the extra gravity is a peripheral baryonic overdensity.

15.6 Observational Hooks and Discriminants: Why This Is Stress-Testable

Distributed forcing implies structured (not arbitrary) phenomenology:

(i) Center offsets

Dynamical centers inferred from kinematics (velocity centroid, circular velocity analysis, dispersion inversion) need not coincide with photometric centers. Offsets should correlate with:

- Configuration asymmetry (deviation from spherical/symmetric distribution),
- Substructure fraction (presence of satellites, merging components),
- Relaxation state (dynamical age, phase-space coherence).

Because low-cost modes reflect the overall geometry and are affected by recent mergers or tidal encounters, center offsets become a sensitive probe of the system's assembly history.

(ii) Multi-lobed morphology

Convergence/potential maps can show multi-blob structure reflecting the excited mode mix rather than a one-to-one mapping between baryonic peaks and gravitational peaks. In the language of Appendix D, if multiple eigenmodes $n = 1, 2, 3, \ldots$ participate with significant amplitudes $|c_n|$, the resulting Φ_{resp} is a superposition of potential patterns, each with its own spatial texture and nodal structure.

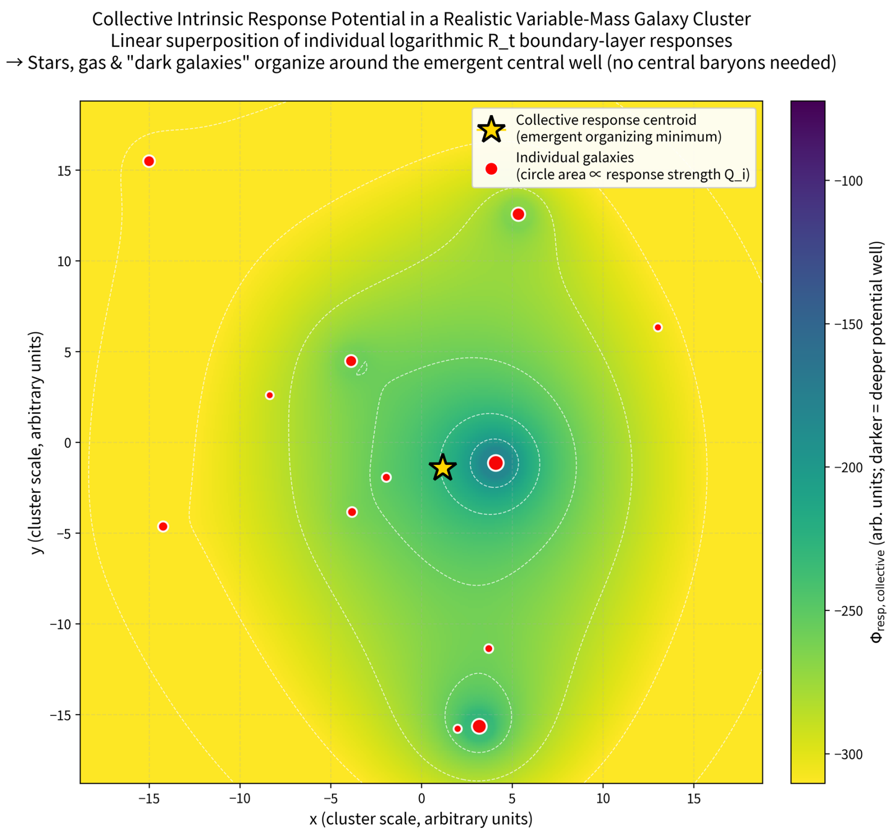

Figure 12: Collective Gravitational Minimum from Distributed Forcing. Schematic representation of the effective potential landscape $\Phi_{\text{eff}}(\mathbf{x}) = \Phi_{\text{bar}}(\mathbf{x}) + \Phi_{\text{resp}}(\mathbf{x})$ in a multi-centered baryonic system. **Red markers:** Individual baryonic concentrations (galaxies or stellar overdensities) with deep local minima from their respective response contributions Φ_i. **Teal depression (broad central bowl):** Collective organizing minimum arising from the superposition of long-wavelength (low-n, fundamental) response eigenmodes. **Gold star:** Location of the global potential minimum, which is generically offset from any single baryonic peak due to the weighted mode geometry. This configuration naturally produces center offsets between dynamical centroids and photometric peaks without requiring exotic dark matter inventories. The organizing center's location depends on the eigenmode profile $\varphi_n(\mathbf{x})$ and the baryonic source distribution, providing a testable prediction for lensing-dynamics consistency studies (Clowe et al., 2006; Euclid Collaboration, 2025).

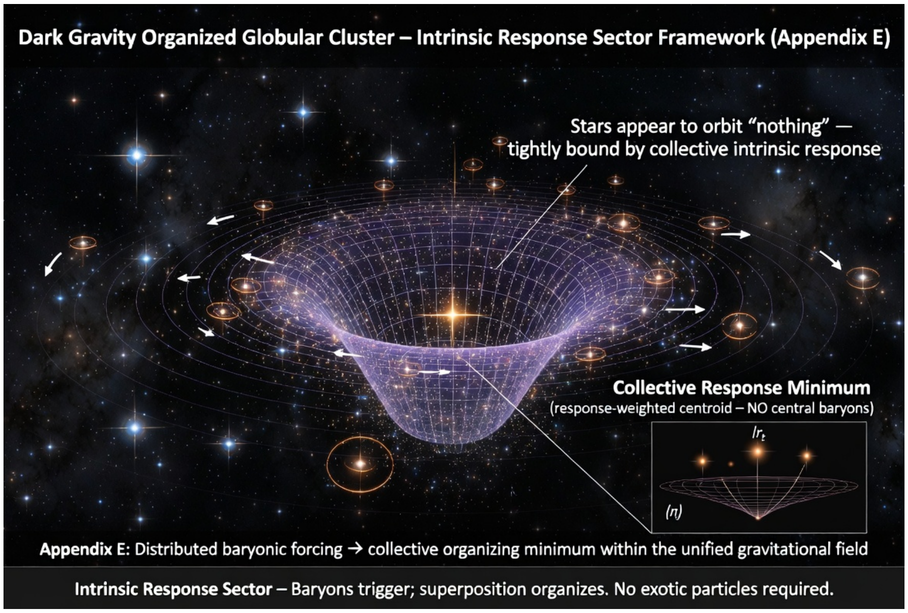

Figure 13: Dark-Gravity Organized Globular Cluster (Schematic). Visualization of a dense stellar system where distributed baryonic drivers (individual stars or stellar subgroups) generate collective intrinsic response contributions whose superposition produces an organizing gravitational minimum. The system exhibits deep local potential wells at each stellar concentration (indicating self-bound objects via their respective R_t response contributions plus baryonic self-gravity), overlaid with a broad collective organizing potential. Outer members orbit this collective structure, appearing to trace "invisible mass" distributions while remaining bound by pure gravitational field organization. This morphology is characteristic of globular clusters and demonstrates how macroscopic response organization can emerge from distributed forcing without requiring exotic dark matter halos.

(iii) Environmental dependence

Because the effective propagation/quenching parameters (α, β in Appendix D.2) can vary with environment, the prevalence and scale of collective minima and offsets should correlate with:

- Tidal field strength (external gravitational stress),
- Local density contrast ($\rho_{\mathrm{local}}/\bar{\rho}$),
- Relaxation measures (two-body encounter rate, Coulomb logarithm).

15.7 Lensing–Dynamics Consistency: The Decisive Rung

Rotation curves constrain primarily the effective matter distribution as experienced by pressure (orbital kinematics), encoded in the dominant term of Φ_{eff}.

Gravitational lensing probes the Weyl combination (metric perturbation), generically:

$$\kappa \propto \Phi_{\mathrm{eff}} + \sum_{\mathrm{slip}} (\text{additional curvature terms}). \tag{10}$$

In a single-field gravitational ontology: the natural expectation is that photons follow the same spacetime geometry as massive tracers. The strongest branch of the framework is **metric-equivalent** behavior in the relevant weak-field regime (minimal gravitational slip, $|\eta_{\mathrm{slip}}| \ll 1$), under which the inferred "dark gravity" contribution lenses as any effective mass distribution would.

Therefore, the strongest discriminant in galaxy groups and clusters is whether the same intrinsic-response organization that shapes dynamics also maps consistently onto lensing (Clowe et al., 2006). If gravitational slip is present in a covariant completion, it must be **controlled** (not opportunistic) and should yield falsifiable relations between dynamical centroids and lensing centroids, including environment- and morphology-dependent signatures that can be confronted directly with group/cluster datasets from *e.g.* Euclid (Euclid Collaboration, 2025), LSST (Ivezić et al., 2019), or dedicated cluster surveys (Mikulski Archive for Space Telescopes (MAST), 2019).

15.8 Summary

Appendix E introduces no new ontology beyond Appendix D. It is the **distributed-forcing extension**: when baryonic driving is multi-centered (groups, clusters, dense stellar systems), the gravitational field's intrinsic response can organize into:

- **Collective minima** and
- **Center offsets**

as global mode-geometry effects operating within the unified gravitational field.

This supplies a coherent, testable alternative to automatically defaulting to an exotic particulate inventory in every "dark gravity" configuration. It identifies lensing–dynamics consistency and environment-dependent centroid structure as the decisive discriminants in the next observational rung.

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